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Internal Control, Cash, and Receivables

**SPOTLIGHT****LEGO**www.lego.com

Founded in 1932 in Denmark, the name LEGO was a mash up of the Danish words “leg godt,” meaning “play well.” Started with wooden toys and then the classic interlocking plastic bricks, it is now one of the most recognizable toys in the world. Many of us have grown up playing with LEGOs at one point in our life. LEGO has launched many new product lines, and more recently we have seen LEGOLAND (theme parks) in a number of countries and even watched LEGO movies. LEGO has its ups and downs, but with sales over \$5 billion and profit of over \$1.3 billion, LEGO is clearly enjoying a surge in popularity despite challenges from digital toys and games. ●

Tpfeller/Shutterstock



The Danish Krone, abbreviated kr, is the official currency of Denmark, Greenland, and the Faroe Islands.

LEGO's total current assets at the end of 2015 were 16,653 million kr. Given LEGO's wide product range, you might think that inventories are the company's largest current asset. Actually, as you can see from its Balance Sheet below, LEGO's largest current asset is its receivables (6,410 and 5,852 million kr). LEGO's cash and receivables total 13,473 kr and represent over 80% of its total current assets. How does LEGO protect and account for these important assets?

A1			
	A	B	C
1	LEGO Group Consolidated Balance Sheet	At December 31	
2	(Adapted, in millions of kr)	2015	2014
3	Cash at banks	kr 1,211	kr 482
4	Inventories	2,747	2,182
5	Trade receivables	6,410	5,891
6	Other receivables	5,852	3,281
7	Prepayments	179	149
8	All other current assets	254	48
9	Total current assets	16,653	12,033
10	Total non-current assets	11,224	9,386
11	Total assets	kr 27,877	kr 21,419
12	Total liabilities	10,126	8,587
13	Total equity	17,751	12,832
14	Total liabilities and equity	kr 27,877	kr 21,419
15			

This chapter begins with a discussion of fraud, its types, and common characteristics. All companies are concerned with the risk of fraud. A basic understanding of what it is would be useful in helping companies safeguard against fraud.

We then discuss internal controls, which are the primary means by which fraud as well as unintentional financial statement errors are prevented. We also discuss how to account for cash and receivables. As cash is probably the asset that is most susceptible to misappropriation through fraud, we will apply internal control concepts on cash receipt and cash disbursement processes.

LEARNING OBJECTIVES

- 1 Understand** the role of internal controls and corporate governance
- 2 Apply** internal controls over cash receipts and cash payments

- 3 Prepare and use** a bank reconciliation
- 4 Account** for receivables and its potential impairment
- 5 Evaluate** a company's ability to collect receivables

UNDERSTAND THE ROLE OF INTERNAL CONTROLS AND CORPORATE GOVERNANCE

Fraud and Its Impact

Fraud is an intentional misrepresentation of facts, made for the purpose of persuading another party to act in a way that causes injury or damage to that party. You may have also heard of other “famous” frauds in the last decade: WorldCom, Enron, Parmalat, Toshiba, Olympus, and unfortunately, the list seems to go on. A more recent financial statement fraud was that of Satyam Computer Services. Headquartered in Hyderabad (India), Satyam was one of the biggest players in the information technology consultancy, integration, and outsourcing business in the world with clients across the globe, including Nestlé, General Motors, and General Electric. In fact, about 185 Fortune 500 companies were clients of Satyam, which employed over 53,000 people. Ramalinga Raju, former Chairman of Satyam Computer, intentionally misstated cash, revenue, receivables, and payables in Satyam’s financial statements. His actions caused shareholders of Satyam Computer Services to lose billions of dollars. In early 2009, Ramalinga Raju’s resignation letter (and admission of wrongdoing) shocked the business world. An excerpt is reproduced below, with references to Indian numerical conventions of *crores* (units of ten millions) and *lakhs* (units of one hundred thousands) converted to millions or billions as appropriate.

1 Understand the role of internal controls and corporate governance

The Indian rupee is the official currency of the Republic of India, abbreviated ₹.

January 7, 2009
To the Board of Directors
Satyam Computer Services Ltd.

Dear Board Members,

It is with deep regret, and tremendous burden that I am carrying on my conscience, that I would like to bring the following facts to your notice:

1. The Balance Sheet carries as of September 30, 2008:
 - a. Inflated (non-existent) cash and bank balances of ₹50.40 billion (about \$1.04 billion) as against ₹53.61 billion reflected in the books.
 - b. An accrued interest of ₹3.76 billion which is non-existent.
 - c. An understated liability of ₹12.30 billion on account of funds arranged by me.
 - d. An overstated debtors’ position of ₹4.90 billion (as against ₹26.51 billion reflected in the books).
2. For the September quarter (Q2) we reported a revenue of ₹27.00 billion and an operating margin of ₹6.49 billion (24% of revenues) as against the actual revenues of ₹21.12 billion and an actual operating margin of 610 million rupees (3% of revenues). This has resulted in artificial cash and bank balances going up by ₹5.88 billion in Q2 alone.

The gap in the Balance Sheet has arisen purely on account of inflated profits over a period of last several years (limited only to Satyam standalone, books of subsidiaries reflecting true performance). What started as a marginal gap between actual operating profit and the one reflected in the books of accounts continued to grow over the years. It has attained unmanageable proportions as the size of company operations grew significantly (annualized revenue run rate of ₹112.76 billion in the September quarter, 2008, and official reserves of ₹83.92 billion). The differential in the real

continued on the following page

profits and the one reflected in the books was further accentuated by the fact that the company had to carry additional resources and assets to justify higher level of operations—thereby significantly increasing the costs.

Every attempt made to eliminate the gap failed. As the promoters held a small percentage of equity, the concern was that poor performance would result in a take-over, thereby exposing the gap. It was like riding a tiger, not knowing how to get off without being eaten. Having put these facts before you, I leave it to the wisdom of the board to take the matters forward.

Under the circumstances, I am tendering my resignation as the chairman of Satyam and shall continue in this position only till such time the current board is expanded. My continuance is just to ensure enhancement of the board over the next several days or as early as possible.

I am now prepared to subject myself to the laws of the land and face consequences thereof.

B. Ramalinga Raju
Chairman, Satyam Computer Services Ltd.

Source: Satyam Computer Services Ltd.

What Ramalinga Raju admitted can be summarized as follows.

A1			
	A	B	C
1	Satyam Computer Services Selected Financial Statement Data Income Statement for Quarter Ended September 30, 2008		
2	(Adapted, in crores/ten million ₹)	Reported	Admitted
3	Sales	₹ 2,701	₹ 2,112
4	Gross profit	724	
5	Operating profit	649	61
6			
7	Balance Sheet at Sept. 30, 2008	Reported	Admitted
8	Cash	5,361	321
9	Accounts receivable net	2,651	490
10	Other receivables	676	300
11	Total current assets	9,065	
12	Property, plant, and equipment net	1,434	
13	Goodwill	446	
14	Total non-current assets	1,996	
15	Total assets	11,062	
16	Accounts payable	1,379	
17	Other current liabilities	823	₹ 2,053
18	Total current liabilities	2,202	
19	Long-term debt	488	
20	Total non-current liabilities	488	
21	Total liabilities	2,690	
22	Total equity	8,372	
23	Total liabilities and equity	₹ 11,062	
24			

*Adapted from various sources by the author.

How can a company with such a supposedly sterling reputation commit such a fraud? One billion U.S. dollars in cash unaccounted for, operating profit inflated by over 10 times, receivables overstated by over 500%. Satyam's shares went into freefall immediately following the news, dropping by about 80%, as investors around the world thrashed the company's shares. *Reuters* called it "India's Enron."

Fraud can happen to any business, and businesses spend much time and effort in preventing, detecting, and correcting fraudulent activities. PricewaterhouseCooper's Global Economic Crime Survey 2016 showed that fraud is a global problem, not confined to just developing countries. Globally, more than one-third of the respondents reported that they have experienced some form of frauds.

In its 2016 survey titled *Report to the Nations on Occupational Fraud and Abuse*,¹ the Association for Certified Fraud Examiners (ACFE) revealed the following:

- A typical organization loses 5% of its revenue each year to fraud. Applied to the 2011 gross world product, this translates to a projected annual fraud loss of over \$3.7 trillion. If you apply this 5% to your national GDP, you can imagine potentially how much is lost due to fraudulent activities.
- The median loss in occupational fraud case is \$150,000, with 23.2% of reported cases causing losses of \$1 million or more.
- Asset misappropriation was by far the most common form of occupational fraud, followed by corruption cases and financial statement fraud. In terms of impact, it is the reverse, i.e., financial fraud cases, on average, caused the greatest amount of losses.
- The longer a fraud lasted, the greater the financial damage it caused. A majority of fraudsters try to conceal their illegal activities by creating fake documents or altering physical documents.
- Close to 80% of fraudsters exhibit one or more behavioral red flags, including (a) living beyond one's means, (b) experiencing financial difficulties, (c) having unusually close associations with vendors or customers, or (d) a general "wheeler-dealer" attitude involving unscrupulous behavior.

What are the most common types of fraud? What causes fraud? What can be done to prevent it? There are many types of fraud. Some of the most common types are insurance fraud, forgery, credit card fraud, and identity theft. The two most common types of fraud that impact businesses' financial statements are

1. **Misappropriation of assets.** *This type of fraud is committed by employees of an entity who steal money from the company and cover it up through erroneous entries in the books. Other examples of asset misappropriation include employee theft of inventory, bribery or kickback schemes in the purchasing function, or employee overstatement of expense reimbursement requests.*
2. **Fraudulent financial reporting.** *This type of fraud is committed by company managers who make false and misleading entries in the books, making financial results of the company appear to be better than they actually are. This may be done through aggressive revenue recognition or fictitious sales, using inappropriate accounting treatment and estimates that do not represent the underlying economic phenomenon. The purpose of this type of fraud is to deceive investors and creditors into investing or loaning money to the company that they might not otherwise have invested in or loaned money to.*

Both of these types of fraud involve making false or misleading entries in the books of the company. We call this *cooking the books*. Of these two types, asset misappropriation is the most common, but fraudulent financial reporting is by far the most expensive. The two most notorious financial reporting frauds involved Enron Corporation in 2001 and WorldCom, Inc., in 2002. These two scandals alone rocked the U.S. economy and impacted financial markets across the world. Enron (discussed in Chapter 8) committed fraudulent financial reporting by overstating profits through bogus sales of non-existent assets with inflated values. When Enron's banks found out, they stopped loaning the company money to operate, causing it to go out of business almost overnight.

WorldCom (discussed in Chapter 7) reported expenses as assets and overstated both profits and assets. The company's internal auditor blew the whistle on WorldCom, resulting in the company's eventual collapse. Waste Management (also in Chapter 7) manipulated accounting

¹Association for Certified Fraud Examiners. *Report to the Nations on Occupational Fraud and Abuse*. Austin, TX: ACFE, 2016. <http://www.acfe.com/rtnn2016/docs/2016-report-to-the-nations.pdf>.

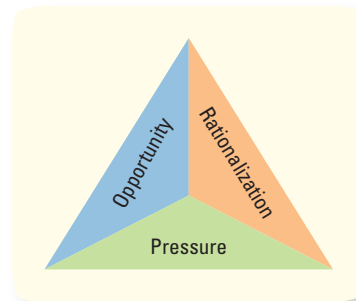
estimates to meet earning projections. Sadly, the same international accounting firm, Arthur Andersen, had audited both companies' financial statements. Because of these and other failed audits, the once mighty firm of Arthur Andersen was forced to close its doors in 2002.

More recently, in one of the biggest ever accounting scandals in Japanese corporate history, Toshiba senior management overstated operating profits by over \$1.2 billion. This was committed over a number of years, by a corporate culture that perpetuates the achievement of sales and revenue targets above ethics (and accounting standards). Employees were pressured by top management into unethical accounting practices, including the deferment of expenses into future years to meet revenue targets.

Each of these frauds, from WorldCom, Waste Management, and Enron to Satyam, involved losses of billions of dollars and thousands of jobs when the companies went out of business or were bought out by other companies. Widespread media coverage sparked adverse market reaction, loss of confidence in the financial reporting system, and losses through declines in share values that ran in the trillions of dollars! We will discuss some of these cases throughout the text as examples of how accounting principles were deliberately misapplied, through cooking the books, in environments characterized by *weak internal controls*.

Exhibit 5-1 explains in graphic form the elements that make up virtually every fraud. We call it the **fraud triangle**.

Exhibit 5-1 | The Fraud Triangle



The first element in the fraud triangle is *pressure*. This usually results from either financial or personal pressure. Being saddled with debts, living beyond one's means, being greedy, and having unreasonable expectations to consistently top a previous year's results create pressure to commit fraud (by the perpetrator). Sometimes it is a matter of just never having enough (because some people who commit fraud are already rich by most people's standards). Sometimes it is more about psychological satisfaction, proving that one can "beat the system." Other times, the perpetrator of the fraud might have a legitimate financial need, such as a medical emergency, but he or she uses illegitimate means to meet that need. A recent article in the *Wall Street Journal* indicated that theft by employees was on the rise due to economic hard times. In any case, the prevailing attitude on the part of the perpetrator is, "I need it, I want it, so I'm going to do whatever I have to do to get it."

The second element in the fraud triangle is *opportunity*. Opportunities to commit fraud usually arise through weak internal controls. It might be a breakdown in a key element of controls, such as *improper segregation of duties* and/or *improper access to assets or computer systems*. Or it might result from a weak *control environment*, such as a domineering CEO, a weak or conflicted board of directors, or lax ethical practices, allowing top management to override whatever controls the company has placed in operation for other transactions.

The third element in the triangle is *rationalization*. The perpetrator engages in self-justification for his or her actions through distorted thinking, such as: "I deserve this because I have done so much for the company"; "Nobody treats me fairly"; "No one will ever know"; "Just this once, I won't let it happen again"; or "Everyone else is doing it."

All three elements of the fraud triangle must be present and take over your decision-making process for fraud to occur. Thieves and robbers have no problems with rationalization or with motives; all they need is opportunity. You, on the other hand, may have the opportunity to steal money or property without people knowing, you may even have a great need for them, but if your

moral and ethical values are strong, you will not rationalize the action and will walk away from the opportunity to steal.

Ethics in Business and Accounting Decisions

As we pointed out in our decision model for making ethical accounting and business judgments (introduced in Chapter 1), the decision to engage in fraud is an act with economic, legal, and ethical implications. The perpetrators of fraud usually do so for their own short-term *economic gain*, while others incur *economic losses* that may far outstrip the gains of the fraudsters. Moreover, fraud is defined by state, federal, and international law as *illegal*. Those who are caught and found guilty of fraud ultimately face penalties which include imprisonment, fines, and monetary damages. Finally, from an *ethical* standpoint, fraud violates the rights of many for the temporary betterment of a few, and for the ultimate betterment of no one. At the end of the day, everyone loses! **Fraud is the ultimate unethical act in business!**

Internal Control

The primary way that fraud, as well as unintentional errors, are prevented, detected, or corrected in an organization is through a proper system of internal control. **Internal control** is a plan of organization and a system of procedures, implemented by company management and the board of directors, designed to accomplish the following five objectives:

1. *Safeguard assets.* A company must safeguard its assets against waste, inefficiency, and fraud. If management fails to safeguard assets such as cash or inventory, those assets will slip away. Most retailers would put in some physical control over their merchandise inventory to prevent shoplifting or employee theft.
2. *Encourage employees to follow company policy.* Everyone in an organization—managers and employees—needs to work toward the same goals. A proper system of controls provides clear policies that result in fair treatment of both customers and employees.
3. *Promote operational efficiency.* Companies cannot afford to waste resources. They work hard to make a sale, and they don't want to waste any of the benefits. If the company can buy something for \$30, why pay \$35? Effective controls minimize waste, which lowers costs and increases profits.
4. *Ensure accurate, reliable accounting records.* Accurate records are essential. Without proper controls, records may be unreliable, making it impossible to tell which part of the business is profitable and which part needs improvement. A business could be losing money on every product it sells—unless it keeps accurate records for the cost of its products.
5. *Comply with legal requirements.* Companies, like people, are subject to laws, such as those of regulatory agencies like the Securities and Exchange Commission or SEC (in the United States); stock exchanges; tax authorities; and state, local, and international governing bodies. When companies disobey the law, they are subject to fines, or, in extreme cases, their top executives may even go to prison. Effective internal controls help ensure compliance with the law and help avoid legal difficulties.

How critical are internal controls? They're so important that in some jurisdictions, boards of directors and/or auditors are required to examine and assess the effectiveness of internal controls of publicly listed companies. For example, BASF, our spotlight company in Chapter 4, has this to say about internal controls (emphasis added):

ADAPTED EXCERPTS FROM BASF'S ANNUAL REPORT

Statement of the Board of Executive Directors (partial):

We have established effective internal control and steering systems in order to ensure that the BASF Group's Consolidated Financial Statements and Management's Report comply with applicable accounting rules and to ensure proper corporate

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reporting. The risk management system we have set up is designed such that the Board of Executive Directors can identify material risks early on and take appropriate defensive measures as necessary. The reliability and effectiveness of the internal control and risk management system are continually audited throughout the Group by our internal audit department.

Management Report on Internal Control System (Partial):

Employees involved in the accounting and reporting process meet the qualitative requirements and participate in training on a regular basis. There is a clear assignment of responsibilities between the specialist units, companies and regional service units involved. We strictly adhere to the principles of segregation of duties and dual control, or the “four-eyes principle.” Complex actuarial reports and evaluations are produced by specialized service providers or specially qualified employees.

An internal control system for financial reporting continuously monitors these principles. To this end, methods are provided for the structured and Group-wide uniform evaluation of the internal control system in financial reporting. The significant risks for the BASF Group regarding a reliable control environment for proper financial reporting are reviewed and updated on an annual basis.

Source: BASF 2015 Annual Report, pages 155 and 115

One of the most well-known examples of this requirement is the Sarbanes-Oxley Act (SOX) in the United States. While this is a U.S. regulation, as you will read later, many jurisdictions around the world have enacted similar requirements.

The Sarbanes-Oxley Act (SOX)

As the Enron and WorldCom scandals unfolded, many people asked, “How can these things happen? If such large companies that we have trusted commit such acts, how can we trust any company to be telling the truth in its financial statements? Where were the auditors?” To address public concerns, the U.S. Congress passed the Sarbanes-Oxley Act of 2002 (SOX). SOX revamped corporate governance in the United States and profoundly affected the way that accounting and auditing are done in public companies. Here are some of the SOX provisions:

1. Public companies must issue an internal control report, and the outside auditor must evaluate and report on the soundness of the company’s internal controls.
2. A new body, the Public Company Accounting Oversight Board, has been created to oversee the audits of public companies.
3. An accounting firm may not both audit a public client and also provide certain consulting services for the same client.
4. Stiff penalties await violators: 25 years in prison for securities fraud; 20 years for an executive making false-sworn statements.

In many other parts of the world, responses to corporate failures, frauds, and scandals resulted in similar increased regulatory requirements. In Japan, “J-SOX” (an unofficial term for parts of the Japan’s Financial Instruments and Exchange Law) was passed in response to corporate scandals such as Kanebo, Livedoor, and Murakami Fund. In China, the China Securities Regulatory Commission, the National Audit Office, the China Banking Regulatory Commission, and the China Insurance Regulatory Commission jointly announced the Basic Standard for Enterprise Internal Control (also nicknamed “Chinese SOX”). Many other countries have also raised the bar in terms of corporate governance practices. The United Kingdom’s Financial Reporting Council recently revised its Combined Code on Corporate Governance. In the European Union, there are various Company Law Directives that regulate company audits, disclosure, and governance.

You may be wondering what all of these internal controls, governance, and risk issues have to do with accounting. Simple, accounting matters! The financial information that an accounting

system produces is crucial information to capital market participants. Accountants (and auditors) thus have a clear role to play to ensure that the financial information is prepared in accordance with applicable rules, regulations, and ethics!

Exhibit 5-2 shows the shield that internal controls provide for an organization. Protected by this shield, which provides protection from fraud, waste, and inefficiency, companies can do business in a trustworthy manner that ensures public confidence, an extremely important element in maintaining the stability of financial markets around the world.

Exhibit 5-2 | The Shield of Internal Control



How does a business achieve good internal controls? The next section will discuss common internal control procedures deployed by business entities. To help you understand the discussion better, we will use the story about EPIC Products, adapted from a case study on a real company, shown in Exhibit 5-3.

Exhibit 5-3 | Cooking the Books: EPIC Products

The following is adapted from a true story:

"I've never been so shocked in my life!" exclaimed Lee Riffe, manager of the EPIC Products office in Palo Alto, California.

"I never thought this could happen to us. We are such a close-knit organization where everyone trusts everyone else. Why, people at EPIC feel like family! I feel betrayed, violated."

Riffe had just returned from the trial of Melissa Price, who had been convicted of embezzling over \$600,000 from EPIC over a six-year period. Price had been one of EPIC's most trusted employees for 10 years. A single mom with two teenage daughters, Price had pulled herself up by her own bootstraps, putting herself through community college where she had obtained an associate's degree in accounting. Riffe had hired her as a part-time bookkeeper at EPIC while Price was in college to help her out. She had done such a good job that when she completed her degree, Riffe asked her to stay on and assigned her the additional role of cashier, in charge of accumulating the daily cash receipts from customers and taking them to the night depository at the bank each day after work. Through the years, he also awarded her what he considered good raises, compensating her at a rate that was generally higher than other employees with her education and experience levels.

Price rapidly became the company's "go-to" financial employee. She was eager to learn, dependable, responsible. In 10 years she never took a day of vacation, choosing instead to take advantage of the company's policy that allowed employees to draw additional compensation for vacation accrued but not taken at the end of each year. Riffe grew to depend on Price more and more each month, as the business grew to serve over 1,000 customers. Price's increased involvement on the financial side of the business freed Riffe to spend his time working on new business, spending less and less time on financial matters. Riffe had noticed that in the past few years, Price had begun to wear better clothes and drive a shiny late-model convertible around town. Both of her teenagers also drove late-model automobiles, and the family had recently moved into a new home in an upscale subdivision of the city. Riffe had been pleased that he had contributed to Price's success. But in recent months, Riffe was becoming worried because, in spite of increasing revenues, the cash balances and cash flows from operations at EPIC had been steadily deteriorating, sometimes causing the company difficulty in paying its bills on time.

Price, on the other hand, had felt underappreciated and underpaid for all of her hard work. Having learned the system well, and observing that no one was monitoring her, Price fell into a simple but deadly trap. As cashier, she was in charge of receiving customer payments that came in by mail. Unknown to Riffe, Price had been lapping accounts receivable, an embezzlement scheme nicknamed “robbing Peter to pay Paul.” Price began by misappropriating (stealing) some of the customers’ checks, endorsing them, and depositing them to her own bank account. To cover up the shortage in a particular customer’s account, Price would apply the collections received later from another customer’s account. She would do this just before the monthly statements were mailed to the first customer, so that the customer wouldn’t notice when he or she received the statement that someone else’s payment was being applied to the amount owed to EPIC. Of course, this left the second customer’s account short, so Price had to misapply the collection from a third customer to straighten out the discrepancy in the second customer’s account. She did this for many customers, over a period of many months, boldly stealing more and more each month. With unlimited access to both cash and customer accounts, and with careful planning and constant diligence, Price became very proficient at juggling entries in the books to keep anyone from discovering her scheme. This embezzlement went on for six years, allowing Price to misappropriate \$622,000 from the company. The customer accounts that were misstated due to the fraud eventually had to be written off.

What tipped off Riffe to the embezzlement? Price was involved in an automobile accident and couldn’t work for two weeks. The employee covering for Price was swamped with telephone calls from customers wanting to discuss unexplained differences in their billing statements for amounts they could prove had been paid. The ensuing investigation pointed straight to Price, and Riffe turned the case over to the authorities.

Due to Price’s scheme, the company had been cheated of \$622,000 over several years that it could have used to buy new equipment, expand operations, or pay off debts. EPIC Products has now revamped its internal controls. The company has hired a separate person, with no access to cash, to keep customer accounts receivable records. The company now uses a lock-box system for all checks received by mail. They are sent to EPIC’s bank lock box, where they are gathered by a bank employee and immediately deposited. The remittance advices accompanying the checks are electronically scanned and forwarded to EPIC’s accounts receivable bookkeeper where they are used as the source documents for posting amounts collected from customers. A summary of cash received goes to Riffe, who reviews it for reasonableness and compares it with the daily bank deposit total. Another employee, who has neither cash handling nor customer bookkeeping responsibilities, reconciles EPIC’s monthly bank statement, and reconciles the total cash deposited per the daily listings with the total credits to customer accounts receivable. Now Riffe requires every employee to take time off for earned vacation, and he rotates other employees through those positions while those employees are away.

Internal Control Procedures

Whether it is Alibaba, Daimler, Richemont, BASF, or LEGO, every major class of transactions needs to be protected by the following *internal control procedures*.

Smart Hiring Practices and Separation of Duties. In a business with good internal controls, no important duty is overlooked. Each person in the information chain is important. The chain should start with hiring. Background checks should be conducted on job applicants. Proper training and supervision, as well as paying competitive salaries, help ensure that all employees are sufficiently competent for their jobs. Employee responsibilities should be clearly laid out in position descriptions. For example, the **treasurer’s** department should be in charge of cash handling, as well as signing and approving checks. Warehouse personnel should be in charge of storing and keeping track of inventory. With clearly assigned responsibilities, all important jobs get done.

In processing transactions, smart management *separates three key duties: asset handling, record keeping, and transaction approval*. For example, in the case of EPIC Products, separation of the duties of cash handling from record keeping for customer accounts receivable would have removed Melissa Price’s incentive to engage in fraud, because it would have made it impossible for her to have lapped accounts receivable if another employee had been keeping the books. Ideally, someone else should also review customer accounts for collectability and be in charge of writing them off if they become completely uncollectible.

The accounting department should be completely separate from the operating departments, such as production and sales. What would happen if sales personnel, who were compensated based on a percentage of the amount of sales they made, approved the company’s sales transactions to

customers? Sales figures could be inflated and might not reflect the eventual amount collected from customers.

At all costs, accountants must not handle cash, and cash handlers must not have access to the accounting records. If one employee has both cash-handling and accounting duties, that person can steal cash and conceal the theft. This is what happened at EPIC Products.

For companies that are *too small* to hire separate persons to do all of these functions, the key to good internal control is *getting the owner involved*, usually by approving all large transactions, making bank deposits, or reconciling the monthly bank account.

Comparisons and Compliance Monitoring. No person or department should be able to completely process a transaction from beginning to end without being cross-checked by another person or department. For example, separate divisions of the treasurer's department should be responsible for depositing daily cash receipts in the bank. The **controller's** department should be responsible for recording customer collections to individual customer accounts receivable. A third employee (perhaps the person in the controller's department who reconciles the bank statement) should compare the treasurer department's daily records of cash deposited with the total collections posted to individual customer accounts by the accounting department.

One of the most effective tools for monitoring compliance with management's policies is the use of **operating budgets** and **cash budgets**. A **budget** is a quantitative financial plan that helps control day-to-day management activities. Management may prepare these budgets on a yearly, quarterly, monthly, or on a more frequent basis. Operating budgets are budgets of future periods' net income. They are prepared by line item of the Income Statement. Cash budgets, discussed in depth later in this chapter, are budgets of future periods' cash receipts and cash disbursements. Often these budgets are "rolling," being constantly updated by adding a time period a year away while dropping the time period that has just passed. Computer systems are programmed to prepare exception reports for data that are out of line with expectations. This data can include variances for each account from budgeted amounts. Department managers are required to explain the variances, and to take corrective actions in their operating plans to keep the budgets in line with expectations. This is an example of the use of **exception reporting**.

To validate the accounting records and monitor compliance with company policies, most companies have an audit. An **audit** is an examination of the company's financial statements and its accounting system, including its controls.

Audits can be internal or external. *Internal auditors* are employees of the business. They ensure that employees are following company policies and operations are running efficiently. Internal auditors also determine whether the company is following legal requirements.

External auditors are completely independent of the business. They are hired to determine whether or not the company's financial statements agree with generally accepted accounting principles. Auditors examine the client's financial statements and the underlying transactions in order to form a professional opinion on the accuracy and reliability of the company's financial statements.

Adequate Records. *Accounting records* provide the details of business transactions. The general rule is that all major groups of transactions should be supported by either hard copy documents or electronic records. Examples of documents include sales invoices, shipping records, customer remittance advices, purchase orders, vendor invoices, receiving reports, and canceled (i.e., paid) checks. Documents should be prenumbered to assure completeness of processing and proper transaction cutoff, and to prevent theft and inefficiency. A gap in the numbered document sequence draws attention to the possibility that transactions might have been omitted from processing.

Limited Access. To complement segregation of duties, company policy should limit access to assets only to those persons or departments that have custodial responsibilities. For example, access to cash should be limited to persons in the treasurer's department. Cash receipts might be processed through a **lock-box system**. Access to inventory should be limited to persons in the company warehouse where inventories are stored, or to persons in the shipping and receiving functions. Likewise, the company should limit access to records to those persons who have record-keeping responsibilities. All manual records of the business should be protected by lock and key, and electronic records should be protected by passwords. Only authorized persons should have

access to certain records. Individual computers in the business should be protected by user identification and password. Electronic data files should be encrypted (processed through a special code) to prevent their recognition if accessed by a “hacker” or other unauthorized person.

Proper Approvals. No transaction should be processed without management’s general or specific approval. The bigger the transaction, the more specific approval it should have. For individual small transactions, management might delegate approval to a specific department. For example:

- Sales to customers on account should all be approved by a separate *credit department* that reviews all customers for creditworthiness before goods are shipped to customers on credit. This helps assure that the company doesn’t make sales to customers who cannot afford to pay their bills.
- Purchases of all items on credit should be approved by a separate *purchasing department* that specializes in that function. Among other things, a purchasing department should only buy from approved vendors, on the basis of competitive bids, to assure that the company gets the highest quality products for the most competitive prices.
- All personnel decisions, including hiring, firing, and adjusting pay, should be handled by a separate *human resources (HR) department* that specializes in personnel-related matters.

Very large (material) transactions should generally be approved by top management, and may even require the board of directors’ approval.

Information Technology. Accounting systems are relying less on manual procedures and more on information technology (IT) than ever before for record keeping, asset handling, approval, and monitoring, as well as physically safeguarding the assets. For example, retailers such as Marks and Spencer and Woolworths control inventory by attaching an *electronic sensor* to merchandise. The cashier must remove the sensor before the customer can walk out of the store. If a customer tries to leave the store with the sensor attached, an alarm sounds. According to *Checkpoint Systems*, these devices reduce theft by as much as 50%. *Bar codes* speed checkout at retail stores, performing multiple operations in a single step (see Exhibit 5-4). When the sales associate scans the merchandise at the register, the computer records the sale, removes the item from inventory, and computes the amount of cash tendered.

When a company employs sophisticated IT, the basic attributes of internal control do not change, but the procedures by which these attributes are implemented change substantially. For example, segregation of duties is often accomplished by separating mainframe computer departments from other user departments (i.e., controller, sales, purchasing, receiving, credit, HR, treasurer) and restricting access to the IT department only to authorized personnel. Within the computer department, programmers should be separated from computer operators and data librarians. Access to sensitive data files is protected by password and data encryption. Electronic

Exhibit 5-4 | Bar Codes Speed Up the Checkout Process



Vartanov Anatoly/Shutterstock

records must be saved routinely, or they might be written over or erased. Comparisons of data (such as cash receipts with total credits to customer accounts) that might otherwise be done by hand are performed by the computer. Computers can help monitor inventory levels by item, generating a purchase order for the inventory when it reaches a certain level.

The use of computers has the advantage of speed and accuracy (when programmed correctly). However, a computer that is *not* programmed correctly can corrupt *all* the data, making it unusable. It is, therefore, important to hire experienced and competent people to run the IT department, to restrict access to sensitive data and the IT department only to authorized personnel, to check data entered into and retrieved from the computer for accuracy and completeness, and to test and retest programs on a regular basis to assure data integrity and accuracy.

Internal Controls for e-Commerce. As businesses and consumers conduct more transactions over the Internet, e-commerce creates its own risks. Hackers may gain access to confidential information such as account numbers and passwords, resulting in loss of confidence in the business. E-commerce pitfalls include the following:

- **Stolen Credit Card Numbers.** Suppose you buy music from iTunes. To make the purchase, your credit card number must travel through cyberspace. Wireless networks (Wi-Fi) are creating new security hazards. Amateur hacker Carlos Salgado, Jr., used his home computer to steal 100,000 credit card numbers with a combined limit exceeding \$1 billion. Salgado was caught when he tried to sell the numbers to an undercover FBI agent.
- **Computer Viruses and Trojan Horses.** A **computer virus** is a malicious program that (1) enters program code without consent and (2) performs destructive actions in the victim's computer files or programs. A Trojan horse is a malicious computer program that hides inside a legitimate program and works like a virus. Viruses can destroy or alter data, make bogus calculations, and infect files. Most firms have found a virus in their system at some point.
- **Phishing Expeditions.** Thieves phish by creating bogus websites that sound very similar to the original websites to spoof and trick visitors, usually via spam emails, such as ebay-uk.com and BankAmerica.com. The “almost-right” sounding websites attract lots of visitors, and the thieves obtain account numbers and passwords from unsuspecting people. The thieves then use the data for illicit purposes.

The Limitations of Internal Control—Costs and Benefits

Unfortunately, most internal controls can be overcome. Collusion—two or more people working together—can beat internal controls. Consider EPIC Products, discussed in Exhibit 5-3. Even if Riffe were to hire a new person to keep the books, if that person had a relationship with Price and if they conspired with each other, they could design a scheme to lap accounts receivable, the same as Price did, and split the take. Other ways to circumvent a good system of internal controls include management override, human limitations such as fatigue and negligence, and gradual deterioration over time due to neglect. Because of the cost/benefit principle, discussed in the next paragraph, internal controls are not generally designed to detect these types of breakdowns. The best a company can do in this regard is to exercise care in hiring honest persons who have no conflicts of interest with existing employees, and to exercise constant diligence in monitoring the system to assure it continues to work properly.

The stricter the internal control system, the more it costs. An overly complex system of internal control can strangle the business with red tape. How tight should the controls be? Internal controls must be judged in light of their costs and benefits. The following is an example of a good cost/benefit relationship. A part-time security guard at a WalMart store costs about \$28,000 a year. If we can show that, on average, each part-time guard prevents about \$50,000 of theft, then the net saving to WalMart is \$22,000. Most people would say the extra guard is well worth the cost!

APPLY INTERNAL CONTROLS OVER CASH RECEIPTS AND CASH PAYMENTS

2 Apply internal controls over cash receipts and cash payments

Internal Control over Cash Receipts

Cash requires some specific internal controls because cash is relatively easy to steal and it's easy to convert to other forms of wealth. Moreover, all transactions ultimately affect cash. Let's see how to control cash receipts.

All cash receipts should be deposited for safekeeping in the bank—quickly. Companies receive cash over the counter and through the mail. Each source of cash has its own security measures.

Cash Receipts over the Counter. Exhibit 5-5 illustrates the purchase of products in a grocery store. The point-of-sale terminal provides control over the cash receipts, while also recording the sale and relieving inventory for the appropriate cost of the goods sold. Consider a Whole Foods Market store. For each transaction, the Whole Foods sales associate issues a receipt to the customer as proof of purchase. The cash drawer opens when the sales associate enters a transaction, and the machine electronically transmits a record of the sale to the store's main computer. At the end of each shift, the sales associate delivers his or her cash drawer to the office, where it is combined with cash from all other terminals and delivered by armored car to the bank for deposit, as explained in the next section. Later, a separate employee in the accounting department reconciles the electronic record of the sales per terminal to the record of the cash turned in. These measures, coupled with oversight by a manager, discourage theft.

Exhibit 5-5 | Cash Receipts Over the Counter



CandyBox Images/Shutterstock

Point-of-sale terminals also provide effective control over inventory. For example, in a restaurant, these devices track sales by menu item and total sales by cash, type of credit card, gift card redeemed, etc. They create the daily sales journal for that store, which, in turn, interfaces with the general ledger. Managers can use records produced by point-of-sale terminals to check inventory levels and compare them against sales records for accuracy. For example, in a restaurant, an effective way to monitor sales of expensive wine is for a manager to perform a quick count of the bottles on hand at the end of the day and compare it with the count at the end of the

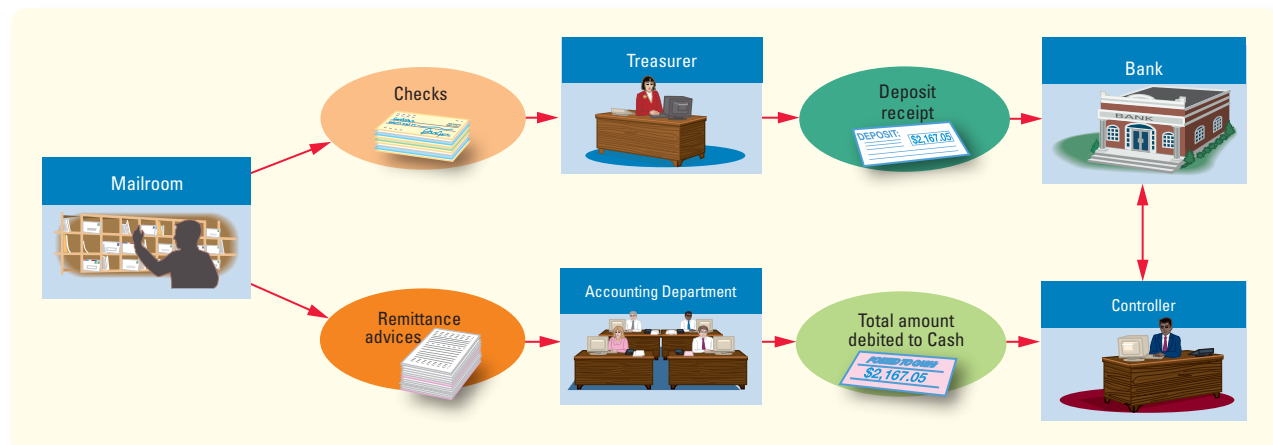
previous day, plus the record of any purchased. The count at the end of the previous day, plus the record of bottles purchased, minus the count at the end of the current day should equal the amount sold as recorded by the point-of-sale terminals in the restaurant.

An effective control for many chain retail businesses, such as restaurants, grocery stores, or clothing stores, to prevent unauthorized access to cash as well as to allow for more efficient management of cash, is the use of “depository bank accounts.” Cash receipts for an individual store are deposited into a local bank account (preferably delivered by armored car for security reasons) on a daily basis. The corporate headquarters arranges for its centralized bank to draft the local depository accounts on a frequent (perhaps daily) basis to get the money concentrated into the company’s centralized account, where it can be used to pay the corporation’s bills. Depository accounts are “one-way” accounts where the local management may only make deposits. They have no authority to write checks on the account or take money out of the store’s account.

Cash Receipts by Mail. Many companies receive cash (in the form of checks or bank orders) by mail, usually from corporate customers and sometimes from retail customers. Exhibit 5-6 shows how companies control cash received by mail. All incoming mail is opened by a mailroom employee. The mailroom then sends all customer checks to the treasurer, who has the cashier deposit the money in the bank. The remittance advices go to the accounting department for journal entries to Cash and customer accounts receivable. As a final step, the controller compares the following records for the day:

- Bank deposit amount from the treasurer
- Debit to Cash from the accounting department

Exhibit 5-6 | Cash Receipts by Mail



The debit to Cash should equal the amount deposited in the bank. All cash receipts are safe in the bank, and the company books are up-to-date.

To prevent unauthorized access to cash, many companies use a bank lock-box system, rather than risk processing checks through the mailroom. Customers send their checks by return mail directly to a post office box controlled by the company’s bank. The bank sends a detailed record of cash received, by customer, to the company for use in posting collections to Accounts Receivable. Internal control is tight because company personnel never touch incoming cash. The lock-box system also gets the cash to the bank in a more timely manner, allowing the company to put the cash to work faster than would be possible if it were processed by the company’s mailroom.

Internal Control over Cash Payments

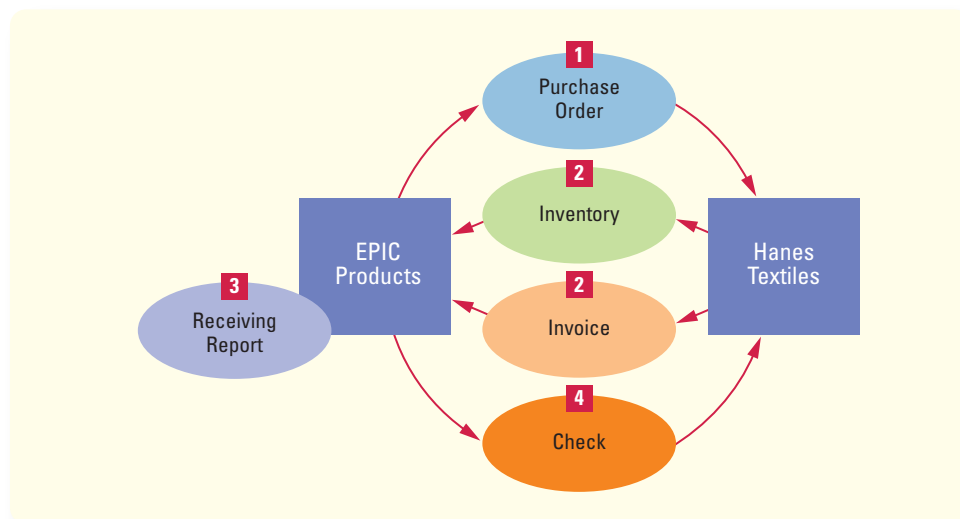
Companies make most payments by check. Let’s see how cash payments by check can be controlled.

Controls over Payments by Check. As we have seen, you need a good separation of duties between (1) operations and (2) writing checks for cash payments. Payment by check is an important internal control, as follows:

- The check provides a record of the payment.
- The check must be signed by an authorized official.
- Before signing the check, the official should study the evidence supporting the payment.

Controls over Purchase and Payment. To illustrate the internal control over cash payments by check, suppose EPIC Products buys some of its inventory from Hanes Textiles. The purchasing and payment process follows these steps, as shown in Exhibit 5-7. Start with the box for EPIC Products on the left side.

Exhibit 5-7 | Cash Payments by Check



1. EPIC Products faxes or e-mails an electronic *purchase order* to Hanes Textiles. EPIC Products says, “Please send us 100 T-shirts.”
2. Hanes Textiles ships the goods and sends an electronic or paper *invoice* back to EPIC Products. Hanes sends the goods.
3. EPIC Products receives the *inventory* and prepares a *receiving report* to list the goods received. EPIC Products receives its T-shirts.
4. After approving all documents, EPIC Products sends a *check* to Hanes, or authorizes an electronic funds transfer (EFT) directly from its bank to Hanes’ bank. By this action, EPIC Products says, “Okay, we’ll pay you.”

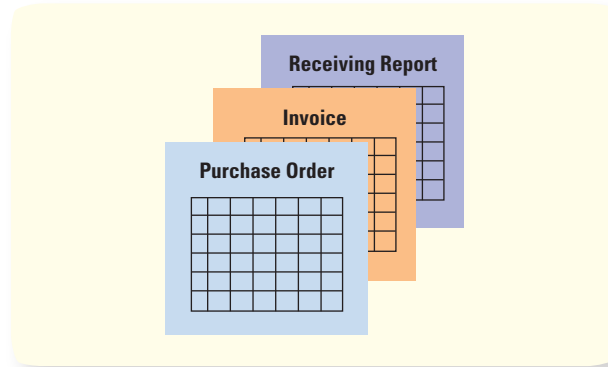
For good internal control, the purchasing agent should neither receive the goods nor approve the payment. If these duties aren’t separated, a purchasing agent can buy goods and have them shipped to his or her home. Or a purchasing agent can spend too much on purchases, approve the payment, and split the excess with the supplier. To avoid these problems, companies split the following duties among different employees:

- Purchasing goods
- Receiving goods
- Approving and paying for goods

Exhibit 5-8 shows EPIC Products' payment packet of documents. Before signing the check or approving the EFT, the treasurer's department should examine the packet to prove that all the documents agree. Only then does the company know that

1. it received the goods ordered.
2. it is paying only for the goods received.

Exhibit 5-8 | Payment Packet



After payment, the person in the treasurer's department who has authorized the disbursement stamps the payment packet "paid" or punches a hole through it to prevent it from being submitted a second time. Dishonest people have tried to run a bill through twice for payment. The stamp or hole shows that the bill has been paid. If checks are used, they should then be mailed directly to the payee without being allowed to return to the department that prepared them. To do so would violate separation of the duties of cash handling and record keeping, as well as unauthorized access to cash.

Petty Cash. It would be wasteful to write separate checks for an executive's taxi fare, name tags needed right away, or delivery of a package across town. Therefore, companies keep a **petty cash** fund on hand to pay such minor amounts. The word "petty" means small. That's what petty cash is—a small cash fund kept by a single employee for the purpose of making such on-the-spot minor purchases or reimbursements.

The petty cash fund is opened with a particular amount of cash. A check for that amount is then issued to the custodian of the petty cash fund, who is solely responsible for accounting for it. Assume that on February 28, Cisco Systems, the worldwide leader in networks for the Internet, establishes a petty cash fund of \$500 in a sales department by writing a check to the designated custodian. The custodian of the petty cash fund cashes the check and places \$500 in the fund, which may be a cash box or other device.

For each petty cash payment, the custodian prepares a petty cash voucher to list the item purchased. The sum of the cash in the petty cash fund plus the total of the paid vouchers in the cash box should equal the opening balance at all times—in this case, \$500. The Petty Cash account keeps its \$500 balance at all times. Maintaining the Petty Cash account at this balance, supported by the fund (cash plus vouchers), is how an **imprest system** works. The control feature is that it clearly identifies the amount for which the custodian is responsible.

PREPARE AND USE A BANK RECONCILIATION

Cash is the most liquid asset because it is the medium of exchange for all economies in the world. Cash is easy to conceal and relatively easy to steal. As a result, most businesses create specific controls for cash.

Keeping cash in a bank account helps control cash because banks have established practices for safeguarding customers' money. The documents used to control a bank account include the following:

- signature card
- deposit ticket
- check
- bank statement
- bank reconciliation

Signature Card. Banks require each authorized person to sign on an account to provide a *signature card*. This helps safeguard against forgery.

Deposit Ticket/Slip. Banks supply standard forms such as *deposit tickets*. The customer fills in the amount of each deposit. As proof of the transaction, the customer keeps a deposit receipt.

Check. To pay cash, the depositor can write a **check**, which tells the bank to pay the designated party a specified amount. There are three parties to a check:

- the maker, or drawer, who signs the check
- the payee, to whom the check is paid
- the bank on which the check is drawn

Exhibit 5-9 shows a check drawn by EPIC Products, the maker. The check has two parts, the check itself and the **remittance advice** below. This optional attachment, which may often be scanned electronically, tells the payee the reason for the payment and is used as the source document for posting to proper accounts.

Exhibit 5-9 | Check with Remittance Advice

The diagram illustrates the components of a check and its associated remittance advice. The check is from BAY AREA NATIONAL BANK, payable to California Office Products for \$320.00. The check number is 338, dated Dec. 28, 20X6. The check is signed by Edward G. Lee, Treasurer, and Horace R. Nash, Vice-President. The remittance advice table shows a payment of \$320.00 on 12/28/X6 for Office Supplies.

Date	Description	Amount
12/28/X6	Office Supplies	\$320.00

Bank Statement. Banks send monthly statements to customers. A **bank statement** reports what the bank did with the customer's cash. The statement shows the account's beginning and ending balances, cash receipts, and payments. Exhibit 5-10 is the December bank statement of the Palo Alto office of EPIC Products.

Exhibit 5-10 | Bank Statement

BAY AREA NATIONAL BANK		BANK STATEMENT			
SOUTH PALO ALTO #136 P.O. BOX 22985 PALO ALTO, CA 94306					
EPIC Products 3814 Glenwood Parkway Palo Alto, CA 94306		ACCOUNT 136-213733 DECEMBER 31, 20X6			
BEGINNING BALANCE	TOTAL DEPOSITS	TOTAL WITHDRAWALS	SERVICE CHARGES		
6,550	4,370	5,000	20		
ENDING BALANCE					
5,900					
TRANSACTIONS					
DEPOSITS		DATE	AMOUNT		
Deposit		12/04	1,150		
Deposit		12/08	190		
EFT—Receipt of cash dividend		12/17	900		
Bank Collection		12/26	2,100		
Interest		12/31	30		
CHARGES		DATE	AMOUNT		
Service Charge		12/31	20		
CHECKS					
Number	Amount	Number	Amount	Number	Amount
307	100	333	150	335	100
332	3,000	334	100	336	1,100
OTHER DEDUCTIONS		DATE	AMOUNT		
NSF		12/04	50		
EFT—Insurance		12/20	400		

Electronic funds transfer (EFT) moves cash by electronic communication. It is cheaper to pay without having to mail a check, so many people pay their mortgage, rent, utilities, and insurance by EFT.

Bank Reconciliation. There are two records of a business's cash:

1. The bank statement, which shows the cash receipts and payments transacted through the bank. In Exhibit 5-10 the bank shows an ending balance of \$5,900 for EPIC Products.
2. The Cash account in the company's general ledger. Exhibit 5-11 shows that EPIC Products' ending cash balance is \$3,340.

Exhibit 5-11 | Cash Records of EPIC Products

General Ledger:

ACCOUNT Cash				
Date	Item	Debit	Credit	Balance
20X6				
Dec 1	Balance			6,550
2	Cash receipt	1,150		7,700
7	Cash receipt	190		7,890
31	Cash payments		6,150	1,740
31	Cash receipt	1,600		3,340

Cash Payments:

A1			
	A	B	C
1	Check No.	Amount	Check No.
2	332	\$ 3,000	337
3	333	510*	338
4	334	100	339
5	335	100	340
6	336	1,100	Total
7			
			\$ 6,150

*Correct amount of check #333 is \$150. See bank reconciliation in Exhibit 5-12 for correction.

The books and the bank statement usually show different cash balances. Differences arise because of a time lag in recording transactions—two examples follow:

- When you write a check, you immediately deduct it in your checkbook. But the bank does not subtract the check from your account until the bank pays the check a few days later, after your payee presents the check to his or her own bank (which may be different from your bank). And you immediately add the cash receipt for all your deposits. But it may take a day or two for the bank to add deposits to your balance.
- Your EFT payments and cash receipts may be directly deposited by your customers and recorded by the bank before you learn of them.

To ensure accurate cash records, you need to update your cash record—either online or after you receive your bank statement. The result of this updating process creates a **bank reconciliation**, which you must prepare. The bank reconciliation explains all differences between your cash records and your bank balance. The person who prepares the bank reconciliation should have no other cash duties. Otherwise, he or she can steal cash and manipulate the reconciliation to conceal the theft.

Preparing the Bank Reconciliation

Here are the items that appear on a bank reconciliation. They all cause differences between the bank balance and the book balance. We call your cash record the “Books.”

Bank Side of the Reconciliation

1. Items to show on the *Bank* side of the bank reconciliation include the following:
 - a. **Deposits in transit** (outstanding deposits). You have recorded these deposits, but the bank has not. Add **deposits in transit** on the bank reconciliation.
 - b. **Outstanding checks**. You have recorded these checks, but the bank has not yet paid them. Subtract outstanding checks.
 - c. **Bank errors**. Correct all bank errors on the Bank side of the reconciliation. For example, the bank may erroneously subtract from your account a check written by someone else.

Book Side of the Reconciliation

2. Items to show on the *Book* side of the bank reconciliation include the following:
 - a. **Bank collections**. Bank collections are cash receipts that the bank has recorded for your account. But you haven’t recorded the cash receipt yet. Many businesses have their customers pay directly to their bank. This is called a *lock-box system* and reduces theft. An example is a bank collecting an account receivable for you. Add bank collections on the bank reconciliation.
 - b. **Electronic funds transfers**. The bank may receive or pay cash on your behalf. An EFT may be a cash receipt or a cash payment. Add EFT receipts and subtract EFT payments.
 - c. **Service charge**. This cash payment is the bank’s fee for processing your transactions. Subtract service charges.
 - d. **Interest revenue on your check account**. On certain types of bank accounts, you earn **interest** if you keep enough cash in your account. The bank statement tells you of this cash receipt. Add interest revenue.
 - e. **Non-sufficient funds (NSF) checks**. These are cash receipts from customers for which there are not sufficient funds in their bank to cover the amount. The NSF checks (sometimes called “bounced” checks) are treated as cash payments on your bank reconciliation. Subtract NSF checks.
 - f. **The cost of printed checks**. This is usually deducted automatically by the bank and is handled like a service charge. Subtract this cost.
 - g. **Book errors**. Correct all book errors on the Book side of the reconciliation. For example, you may have recorded a \$150 check that you wrote as \$510 on your books.

Bank Reconciliation Illustrated. The bank statement in Exhibit 5-10 shows that the December 31 bank balance of EPIC Products is \$5,900 (upper-right corner). However, the company’s Cash account has a balance of \$3,340, as shown in Exhibit 5-11. This situation calls for a bank reconciliation. Exhibit 5-12, panel A, lists the reconciling items for easy reference, and panel B shows the completed reconciliation.

Exhibit 5-12 | Bank Reconciliation**PANEL A—Reconciling Items****Bank side:**

1. Deposit in transit, \$1,600.
2. Bank error: The bank deducted \$100 for a check written by another company. Add \$100 to the bank balance.
3. Outstanding checks—total of \$1,340.

A1	
A	
1	Check No.
2	337
3	338
4	339
5	340
6	
B	
Amount	
\$ 280	
320	
250	
490	

Book side:

4. EFT receipt of your dividend revenue earned on an investment, \$900.
5. Bank collection of your account receivable, \$2,100.
6. Interest revenue earned on your bank balance, \$30.
7. Book error: You recorded check no. 333 for \$510. The amount you actually paid on account was \$150. Add \$360 to your book balance.
8. Bank service charge, \$20.
9. The NSF check from a customer, \$50. Subtract \$50 from your book balance.
10. The EFT payment of insurance expense, \$400.

PANEL B—Bank Reconciliation

A1	
A	
B	
C	
D	
E	
F	
1	EPIC Products
2	Bank Reconciliation
3	December 31, 20X6
4	Bank
5	Balance, December 31
6	Add:
7	1. Deposit in transit
8	2. Correction of bank error
9	
10	
11	
12	Less:
13	3. Outstanding checks
14	No. 337
15	No. 338
16	No. 339
17	No. 340
18	Adjusted bank balance
19	
20	
21	
22	
23	
24	
25	
26	
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Journalizing Transactions from the Bank Reconciliation. The bank reconciliation is an accountant's tool separate from the journals and ledgers. It does *not* account for transactions in the journal. To get the transactions into the accounts, we must make journal entries and post to the ledger. All items on the Book side of the bank reconciliation require journal entries. The bank reconciliation in Exhibit 5-12 requires EPIC Products to make journal entries to bring the Cash account up to date. The numbers in red correspond to the reconciling items listed in Exhibit 5-12, Panel A.

A1				
	A	B	C	E
1	4.	Dec. 31	Cash	900
2			Dividend Revenue	900
3			Receipt of dividend revenue earned on investment.	
4				
5	5.	31	Cash	2,100
6			Accounts Receivable	2,100
7			Account receivable collected by bank.	
8				
9	6.	31	Cash	30
10			Interest Revenue	30
11			Interest earned on bank balance.	
12				
13	7.	31	Cash	360
14			Accounts Payable	360
15			Correction of check no. 333.	
16				
17	8.	31	Bank Service Charges	20
18			Cash	20
19			Bank service charge.	
20				
21	9.	31	Accounts Receivable	50
22			Cash	50
23			NSF check returned by bank.	
24				
25	10.	31	Insurance Expense	400
26			Cash	400
27			Payment of monthly insurance.	
28				

The entry for the NSF check (entry 9) deserves additional explanation. Upon learning that a customer's \$50 check was not honored (i.e., declined) by the customer's bank, Cash must be credited to update the Cash account. This typically happens when there are insufficient funds for the check issued, an unauthorized signature, or other reasons. Unfortunately, there is still a receivable from the customer, so Accounts Receivable must be debited to reinstate the receivable.

Online Banking

Online banking allows you to pay bills and view your account electronically. You don't have to wait until the end of the month to get a bank statement. With online banking, you can reconcile transactions at any time and keep your account current whenever you wish. Exhibit 5-13 shows a sample page from the account history of a bank account.

Exhibit 5-13 Online Banking: Account History

Transaction History for Account # 5401-632-9 as of Close of Business 07/27/20X6				
Account Details				
Current Balance			\$4,136.08	
Date ↓	Description	Withdrawals	Deposits	Balance
Current Balance				\$4,136.08
07/27/X6	DEPOSIT		1,170.35	
07/26/X6	28 DAYS INTEREST		2.26	
07/25/X6	Check #6131 View Image	443.83		
07/24/X6	Check #6130 View Image	401.52		
07/23/X6	EFT PYMT CINGULAR	61.15		
07/22/X6	EFT PYMT CITICARD PAYMENT	3,172.85		
07/20/X6	Check #6127 View Image	550.00		
07/19/X6	Check #6122 View Image	50.00		
07/16/X6	Check #6116 View Image	2,056.75		
07/15/X6	Check #6123 View Image	830.00		
07/13/X6	Check #6124 View Image	150.00		
07/11/X6	ATM 4900 SANGER AVE	200.00		
07/09/X6	Check #6119 View Image	30.00		
07/05/X6	Check #6125 View Image	2,500.00		
07/04/X6	ATM 4900 SANGER AVE	100.00		
07/01/X6	DEPOSIT		9,026.37	

From Federal Deposit Insurance Corporation. Published by Federal Deposit Insurance Corporation.

The account history—like a bank statement—lists deposits, checks, EFT payments, ATM withdrawals, and interest earned on your bank balance. But depending on the bank's online system, the account history may not show your beginning balance, so you can't work from your beginning balance to your ending balance.

» Stop & Think

The bank statement balance is \$4,500 and shows a service charge of \$15, interest earned of \$5, and an NSF check for \$300. Deposits in transit total \$1,200; outstanding checks are \$575. The bookkeeper recorded as \$152 a check of \$125 in payment of an account payable. This created a book error of \$27 (positive amount to correct the error).

- (1) What is the adjusted bank balance?
- (2) What was the book balance of cash before the reconciliation?

Answers:

- (1) \$5,125 (\$4,500 + \$1,200 - \$575).
- (2) \$5,408 (\$5,125 + \$15 - \$5 + \$300 - \$27). The adjusted book and bank balances are the same. The answer can be determined by working backward from the adjusted balance.

Whether it is conventional banking or online banking, the discussions we had earlier are still applicable. Organizations that pay their bills electronically would also need to ensure that there are proper authorization processes and adequate internal control measures in place. Payments from customers are received directly to our bank accounts, minimizing risks of misappropriation. Bank reconciliations would still be needed to ensure that the bank balances agree with our records.

Using the Bank Reconciliation to Control Cash. The bank reconciliation can be a powerful control device. Jonathan Leong is a CPA in Hong Kong. He owns several apartments that are managed by one of his nephews, Alex. Alex signs up tenants, collects the monthly rents, arranges maintenance work, hires and fires employees, writes the checks, and performs the bank reconciliation. In short, he does it all. This concentration of duties in one person is evidence of weak internal control. Alex could be stealing from him, and as a CPA, Jonathan is aware of this possibility.

Jonathan trusts Alex because he is a member of the family. Nevertheless, Jonathan exercises some controls over Alex's management of his apartments. Jonathan periodically drops by the apartments to see whether the maintenance staff are keeping the property in good condition. To control cash, Jonathan occasionally examines the bank reconciliation that Alex has performed. Jonathan would know immediately if Alex were writing checks to himself. By examining the copy of each check issued, Jonathan establishes control over cash payments.

Jonathan has a simple method for controlling cash receipts. He knows the occupancy level of his apartments. He also knows the monthly rent he charges. Jonathan multiplies the number of apartments—say five—by the monthly rent (which averages \$5,000 per unit) to arrive at expected monthly rent revenue of \$25,000. By tracing the \$25,000 revenue to the bank statement, Jonathan can tell if all his rent money went into his bank account. To keep Alex on his toes, Jonathan lets him know that he periodically audits his work.

Control activities such as these are critical. If there are only a few employees, separation of duties may not be feasible. The manager must control operations, or the assets may slip away.

Using a Budget to Manage Cash

As mentioned earlier in the chapter, a budget is a financial plan that helps coordinate business activities. Managers control operations with an operating budget. They also control cash receipts and cash payments, as well as ending cash balances, through use of a cash budget.

For example, how does EPIC Products decide when to invest in new inventory tracking technology? How will EPIC Products decide how much to spend? Will borrowing be needed, or can EPIC Products finance the purchase with internally generated cash? What do ending cash balances need to be in order to provide a “safety margin” so the company won't unexpectedly run out of cash? A cash budget for a business works on roughly the same concept as a personal budget. By what process do you decide how much to spend on your education? On an automobile? On a house? All these decisions depend to some degree on the information that a cash budget provides.

A cash budget helps a company or an individual manage cash by planning receipts and payments during a future period. The company must determine how much cash it will need and then decide whether or not operations will bring in the needed cash. Managers proceed as follows:

1. Start with the entity's cash balance at the beginning of the period. This is the amount left over from the preceding period.
2. Add the budgeted cash receipts and subtract the budgeted cash payments.
3. The beginning balance plus receipts and minus payments equals the expected cash balance at the end of the period.
4. Compare the cash available before new financing to the budgeted cash balance at the end of the period. Managers know the minimum amount of cash they need (the budgeted balance). If the budget shows excess cash, managers can invest the excess. But if the cash available falls below the budgeted balance, the company will need additional financing. The company may need to borrow the shortfall amount. The budget is a valuable tool for helping the company plan for the future.

The budget period can span any length of time—a day, a week, a month, or a year. Exhibit 5-14 shows a cash budget for EPIC Products, Inc., for the year ended December 31, 20X7. Study it carefully, because at some point you will use a cash budget.

Exhibit 5-14 Cash Budget

A1			
	A	B	C
	EPIC Products Cash Budget For the Year Ended December 31, 20X7		
1			
2	Cash balance, December 31, 20X6		\$ 6,260
3	Budgeted cash receipts:		
4	Collections from customers		55,990
5	Dividends on investments		1,200
6	Sale of store fixtures		5,700
7			69,150
8	Budgeted cash payments:		
9	Purchases of inventory	\$ 33,720	
10	Operating expenses	11,530	
11	Expansion of store	12,000	
12	Payment of long-term debt	5,000	
13	Payment of dividends	3,000	65,250
14	Cash available (needed) before new financing		\$ 3,900
15	Budgeted cash balance, December 31, 20X7		(5,000)
16	Cash available for additional investments, or		
17	(New financing needed)		\$ (1,100)
18			

EPIC Products' cash budget in Exhibit 5-14 begins with \$6,260 of cash at the end of the previous year (Line 1). Then add budgeted cash receipts and subtract budgeted payments for the current year. In this case, EPIC Products expects to have \$3,900 of cash available at year-end (Line 10). EPIC Products managers need to maintain a cash balance of at least \$5,000 (line 11). Line 12 shows that EPIC Products must arrange \$1,100 of financing in order to achieve its goals for 20X7.

Reporting Cash on the Balance Sheet

Most companies have numerous bank accounts, but they usually combine all cash amounts into a single total called "Cash and Cash Equivalents." You saw at the start of this chapter LEGO reports cash and cash equivalent of 1,211 million kr on its Balance Sheet. **Cash equivalents** include highly liquid assets such as time deposits, certificates of deposit, and high-grade government securities that are not subjected to any significant change of value. Slightly less liquid than cash, cash equivalents are sufficiently similar to be reported along with cash. We will discuss cash and cash equivalents further in Chapter 11.

Mid-Chapter

Summary Problem

The cash account of Baylor Associates at February 28, 20X7, follows.

Cash				
Feb. 1	Bal. 3,995	Feb. 3		400
6	800	12		3,100
15	1,800	19		1,100
23	1,100	25		500
28	2,400	27		900
Feb. 28	Bal. 4,095			

Baylor Associates received the bank statement on February 28, 20X7 (negative amounts are in parentheses):

A1			
	A	B	C
1	Bank Statement for February 20X7		
2	Beginning balance		\$ 3,995
3	Deposits:		
4	Feb. 7	\$ 800	
5	15	1,800	
6	24	1,100	3,700
7	Checks (total per day):		
8	Feb. 8	\$ 400	
9	16	3,100	
10	23	1,100	(4,600)
11	Other items:		
12	Service charge		(10)
13	NSF check from M. E. Crown		(700)
14	Bank collection of note receivable for the company		1,000
15	EFT—monthly rent expense		(330)
16	Interest revenue earned on account balance		15
17	Ending balance		\$ 3,070
18			

Additional data:

Baylor deposits all cash receipts in the bank and makes all payments by check.

Requirements

1. Prepare the bank reconciliation of Baylor Associates at February 28, 20X7.
2. Journalize the entries based on the bank reconciliation.

Answers

Requirement 1

A1			
	A	B	C
1	Baylor Associates Bank Reconciliation February 28, 20X7		
2	Bank:		
3	Balance, February 28, 20X7		\$ 3,070
4	Add: Deposit of February 28 in transit		2,400
5			5,470
6			
7	Less: Outstanding checks issued on Feb. 25 (\$500)		
8	and Feb. 27 (\$900)		(1,400)
9	Adjusted bank balance, February 28, 20X7		\$ 4,070
10			
11	Books:		
12	Balance, February 28, 20X7		\$ 4,095
13	Add: Bank collection of note receivable		1,000
14	Interest revenue earned on bank balance		15
15			5,110
16	Less: Service charge	\$ 10	
17	NSF check	700	
18	EFT—Rent expense	330	(1,040)
19	Adjusted book balance, February 28, 20X7		\$ 4,070
20			

equal

Requirement 2

A1				
	A	B	C	D
1	Feb 28	Cash	1,000	
2		Note Receivable		1,000
3		<i>Note receivable collected by bank.</i>		
4				
5	28	Cash	15	
6		Interest Revenue		15
7		<i>Interest earned on bank balance.</i>		
8				
9	28	Miscellaneous Expense	10	
10		Cash		10
11		<i>Bank service charge.</i>		
12				
13	28	Accounts Receivable	700	
14		Cash		700
15		<i>NSF check returned by bank.</i>		
16				
17	28	Rent Expense	330	
18		Cash		330
19		<i>Monthly rent expense.</i>		
20				

ACCOUNT FOR RECEIVABLES AND ITS POTENTIAL IMPAIRMENT

4 **Account** for receivables and its potential impairment

Types of Receivables

Receivables are monetary claims against others. Receivables are acquired mainly by selling goods and services (accounts receivable) and by lending money (notes receivable). The journal entries to record the receivables can be shown as follows:

Performing a Service on Account		Lending Money on a Note Receivable	
Accounts Receivable.....	XXX	Note Receivable	XXX
Service Revenue.....	XXX	Cash.....	XXX
<i>Performed a service on account.</i>		<i>Loaned money to another company.</i>	

The two major types of receivables are accounts receivable and notes receivable. A business's *accounts receivables* are the amounts collectible from customers from the sale of goods and services. Accounts receivable, which are typically classified as *current assets*, are sometimes called *trade receivables*, *debtors*, or sometimes just *receivables*. The use of the word "trade" is usually to separate receivables (and payables) arising from the selling of goods or provision of services to customers (and buying of goods or services from suppliers) as opposed to non-trade receivables. LEGO's Balance Sheet at the start of this chapter showed trade and other receivables as one item (line 5) but there is additional information in the notes to the financial statement on the breakdown between trade versus non-trade receivables.

When items are sold on account, the amount recognized as receivable is the fair value of the consideration to be received (recall our discussion in Chapter 3 on revenue recognition). This receivable meets the definition of an asset because it represents future economic benefits the selling entity expects to collect, and it can be measured reliably through contract, invoice, or some other form of sales documentation.

The Accounts Receivable account in the general ledger serves as a *control account* that summarizes the total amount receivable from all customers. Companies also keep a *subsidiary record* of accounts receivable with a separate account for each customer, illustrated as follows:

General Ledger		Accounts Receivable Subsidiary Record	
Accounts Receivable		Brown	
Balance	9,000	Balance	5,000
		FedEx	
		Balance	1,000
		Moody's	
		Balance	3,000
		Total \$9,000	

Notes receivable are more formal contracts than accounts receivable. For a note, the borrower signs a written promise to pay the lender a definite sum at a future date, plus interest. This is why notes are also called promissory notes. The note may require the borrower to pledge *security* for the loan. This means that the borrower gives the lender permission to claim certain assets, called *collateral*, if the borrower fails to pay the amount due. We cover the details of notes receivable later in this chapter.

Other than trade and other receivables, an entity may also have *other receivables*, which include various advances and loans to employees and other parties.

Internal Controls over Cash Collections on Account

Businesses that sell on credit receive most of their cash receipts from collections of accounts receivable. Internal control over collections on account is just as important as cash collection at point of sale. Earlier we discussed control procedures for cash receipts, but another element of internal control deserves emphasis here—the separation of cash-handling and cash-accounting duties. Consider the following case:

Central Paint Company is a small, family-owned business that takes pride in the loyalty of its workers. Most employees have been with Central for 10 or more years. The company makes 90% of its sales on account and receives most of its cash (in the form of checks) by mail.

The office staff consists of a bookkeeper and an office supervisor. The bookkeeper maintains the general ledger and a subsidiary record of individual customer accounts receivable. The bookkeeper also makes the daily bank deposit.

The supervisor prepares monthly financial statements and any special reports the company needs. The supervisor also takes sales orders from customers and serves as office manager.

Can you identify the internal control weakness here? The problem is that the bookkeeper makes the bank deposit. Remember the EPIC Products case (see Exhibit 5-3)? With this cash-handling duty, the bookkeeper could lap accounts receivable. Alternatively, he or she could steal an incoming customer check and write off the customer's account as uncollectible. The customer doesn't complain because the bookkeeper wrote off the customer's account, and Central, therefore, stops pursuing collection.

How can this weakness be corrected? The supervisor—not the bookkeeper—could open incoming mail and make the daily bank deposit. The bookkeeper should *not* be allowed to handle cash. Only the remittance advices should be forwarded to the bookkeeper to credit customer accounts receivable. Removing cash handling from the bookkeeper and keeping the accounts away from the supervisor separate duties and strengthen internal control.

Using a bank lock box achieves the same separation of duties. Customers send their payments directly to Central Paint Company's bank, which records cash as the cash goes into Central's bank account. The bank then forwards the remittance advice to Central's bookkeeper, who credits the customer account. No Central Paint employee even touches incoming cash.

How Do We Manage the Risk of Not Collecting?

In Chapters 1 to 4, we used many different companies to illustrate how to account for a business. All of them hold substantial amounts of receivables from selling goods or providing services on credit.

Selling on credit creates both a benefit and a cost:

- **Benefit:** Customers who cannot pay cash immediately can buy on credit, so sales and profits increase.
- **Cost:** The company cannot collect from some customers. Accountants label this cost **uncollectible-account expense**, **doubtful-account expense**, or **bad-debt expense**.

By selling on credit, companies run the risk of not collecting some receivables. Unfortunately, some customers don't pay their debts. The prospect of failing to collect from a customer provides the biggest challenge in accounting for receivables. The Decision Guidelines address this challenge.



DECISION GUIDELINES

MANAGING AND ACCOUNTING FOR RECEIVABLES

Here are the management and accounting issues a business faces when the company extends credit to customers. For each issue, the Decision Guidelines propose a plan of action. Let's look at a business situation: Suppose you open a health club near your college. Assume you will let customers use the club and bill them for their monthly dues. What challenges will you encounter by extending credit to customers? The main issues in managing receivables, along with plans of action, are:

Issues

1. What are the benefits and costs of extending credit to customers?
2. Extend credit only to creditworthy customers.
3. Separate cash-handling and accounting duties to keep employees from stealing the cash collected from customers.
4. Pursue collection from customers to maximize cash flow.

Plans of Action

1. Benefit: Increase in sales. Cost: Risk of not collecting.
2. Run a credit check on prospective customers.
3. Design the internal control system to separate duties.
4. Keep a close eye on customer pay habits. Send second and third statements to slow-paying customers, if necessary.

The main issues in accounting for receivables, and the related plans of action, are (amounts are assumed):

Issues

1. Measure and report receivables on the Balance Sheet at their net carrying amount, the amount we expect to collect. This is the appropriate amount to report for receivables.
2. Measure and report the expense associated with failure to collect receivables. This expense is called uncollectible-account expense and is reported on the Income Statement.

Plans of Action

Report receivables at their net carrying amount:

Balance Sheet

Receivables.....	\$1,000
Less: Allowance for uncollectibles	(80)
Receivables, net.....	\$ 920

Measure the expense of not collecting from customers:

Income Statement

Sales (or service) revenue.....	\$8,000
Expenses:	
Uncollectible-account expense.....	190

These guidelines lead to our next topic, Accounting for Uncollectible Receivables.

Accounting for Uncollectible Receivables

A company gets an account receivable only when it sells its product or service on credit (on account). You'll recall that the entry to record the earning of revenue on account is (amount assumed):

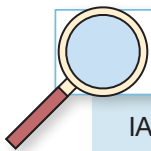
A1				
	A	B	C	D
1		Accounts Receivable	1,000	
2		Sales Revenue (or Service Revenue)		1,000
3		Earned revenue on account.		
4				

Ideally, the company would collect cash for all of its receivables. But unfortunately, this does not happen all the time. How do we account for uncollectible receivables? Uncollectible-account expense is an operating expense along with salaries, depreciation, rent, and utilities. You may also see companies label this as *bad debt expense*, *impairment of receivables expense*, or something similar. To measure uncollectible-account expense, accountants use the allowance method or, in certain limited cases, the direct write-off method (see later in this chapter).

Allowance Method

The best way to provide for bad debts is by the allowance method. *IAS 39—Financial Instruments: Recognition and Measurement* stipulates that loans and receivables, like other financial assets, are impaired if, and only if, there is objective evidence of impairment as a result of one or more “loss events” that occurred after their initial recognition. This includes observable data that comes to the attention of the holder of the loans and receivables about the following loss events:

- Significant financial difficulty of a specific debtor, including the possibility of becoming bankrupt.
- A breach of contract by a specific debtor, such as default or inability to make interest and/or principal payments.
- Adverse changes in the number of delayed payments by debtors in general.
- National or local economic conditions that correlate with defaults by debtors in general (e.g., increase in unemployment rates and other adverse changes in industry conditions that affect debtors).



A Closer Look

IAS 39 is being replaced by *IFRS 9 – Financial Instruments* effective for annual periods beginning January 1, 2018. Some jurisdictions may have a different effective date for the implementation of IFRS 9. With IFRS 9, a new “expected loss impairment model” will result in more timely recognition of impairment losses. Non-financial institutions are allowed to use a simplified matrix model for trade receivables. This model will look similar to the aging of customer balances that will be shown later in this chapter. The specifics of IFRS 9 is beyond the scope of this introductory text, but the basic impairment concepts we discuss here will still be relevant, whether under IAS 39 or IFRS 9.

The allowance method thus records collection losses based on estimates developed from the company's collection experience and information about debtors. LEGO doesn't wait to see which customers will not pay. Instead, LEGO records the estimated amount as Uncollectible-Account Expense and also sets up an *Allowance for Uncollectible Receivables* account. Other titles for this account are *Allowance for Doubtful Receivables* and *Allowance for Receivables Impairment*. This is a contra account to Accounts Receivable. The allowance shows the amount of the receivables the business expects not to collect.

In Chapter 3 we used the Accumulated Depreciation account to show the amount of a PPE's cost that has been expensed—the portion of the asset whose benefits have been allocated to prior periods. Allowance for Doubtful Debt serves a similar purpose for Accounts Receivable. The allowance shows how much of the receivable is unlikely to be collected. You'll find this diagram helpful (amounts are assumed):

Equipment.....	\$100,000	Accounts receivable.....	\$10,000
Less: Accumulated depreciation	(40,000)	Less: Allowance for uncollectible receivables	(900)
Equipment, net.....	60,000	Accounts receivable, net	9,100

Focus on Accounts Receivable. Customers owe this company \$10,000, but it expects to collect only \$9,100. The net receivables is therefore \$9,100. Another way to report these receivables is

Accounts receivable, less allowance of \$900.....	\$9,100
---	---------

You can work backward to determine the full amount of the receivable, \$10,000 (net receivables of \$9,100 plus the allowance of \$900). The Income Statement reports Uncollectible-Account Expense (or receivables impairment expense, bad debt expense, or something similar) among the operating expenses, as follows (using assumed figures):

Equipment.....	\$100,000	Accounts receivable.....	\$10,000
Less: Accumulated depreciation	(40,000)	Less: Allowance for uncollectible accounts	(900)
Equipment, net.....	60,000	Accounts receivable, net	9,100

Income Statement (partial):

Expenses:

Uncollectible-account expense.....	\$2,000
------------------------------------	---------

The best way to estimate uncollectibles uses the company's history of collections from customers and information on "loss events" as suggested by IAS 39 (see earlier). In practice, a popular method for estimating uncollectibles is called aging-of-receivables. The aging method is a Balance-Sheet approach because it focuses on what should be the most relevant and faithful representation of accounts receivable as of the Balance Sheet date. In the aging method, individual receivables from specific customers are analyzed based on how long they have been outstanding. A simplified version of the aging method would simply list the status or age of the receivables, typically classified into age groups such as "Not yet due," "1–30 days overdue," "31–60 days overdue," and "over 60 days overdue." A business entity may estimate that accounts that are not yet due only have a 1% chance of not being collectible, and those that have been overdue for 1–30 days are more likely to be uncollectible, say 5%, and so forth. The allowance for each age group is thus determined *vertically*, and the totals of these allowances will be what the business will need to provide at the year-end. Exhibit 5-15 illustrates a simplified aging method of calculating allowance for uncollectible receivables.

As you can see, the total desired amount of allowance based on our aging analysis is \$1,200 for receivables totaling \$20,000. Whilst we can express this allowance as a percentage of the receivables, i.e., 6% (\$1,200 of \$20,000) of the receivables are expected to be impaired or uncollectible, you don't decide on the 6% first and then calculate the \$1,200. The actual calculation depends on the age groups and the individual percentage uncollectible assigned to each age group.

What is the limitation of this simplified aging method? Well, have a look at Customer C in Exhibit 5-15. Let's say you are now aware of a "loss event" about Customer C, who is very likely to enter bankruptcy. The aging method disregards this fact and only provides an allowance of \$123 ($300 \times 1\% + 200 \times 5\% + 600 \times 15\% + 100 \times 20\%$) when it is possible that all \$1,200 is impaired! If you have assessed each customer's receivable row by row (i.e., *horizontally* rather than by age group *vertically*), you would have taken into account specific information about each customer in your

Exhibit 5-15 Age of Accounts Receivables

Age of Account Receivables					
Receivables	Not yet due	1–30 Days	31–60 Days	Over 60 Days	Total
Customer A	\$400				\$400
Customer B	\$100	\$100			\$200
Customer C	\$300	\$200	\$600	\$100	\$1,200
...	...	...	...	...	...
Totals	\$10,000	\$5,000	\$3,000	\$2,000	\$20,000
Percentage uncollectible	1.0%	5.0%	15.0%	20.0%	
Required allowance.....	<u>\$100</u>	<u>\$250</u>	<u>\$450</u>	<u>\$400</u>	<u>\$1,200</u>

assessment of receivables impairment. For example, you may decide that Customer C's likelihood to be a "loss event" is 80%. You would then set an allowance amounting to \$960 ($\$1,200 \times 80\%$) specifically for Customer C. In practice, a combination of aging and specific customer credit information may be used to determine the required allowance of uncollectible accounts at year-end.

LEGO's accounting policies and notes on receivables are shown below. In total, customers owe LEGO 6,770 million kr. After examining the potential impairment of these receivables, LEGO

ADAPTED EXCERPTS FROM LEGO'S NOTES TO THE FINANCIAL STATEMENTS

Receivables

Trade receivables are initially recognized at fair value and subsequently measured at amortized cost less write down for losses. Allowance for doubtful receivables are made on basis of an objective indication if an individual receivable or a portfolio of receivables are impaired.

Management makes allowance for doubtful trade receivables in anticipation of estimated losses resulting from the subsequent inability of customers to make required payments. Management analyses trade receivables and examines historical bad debt, customer concentrations, customer creditworthiness, and payment history and changes in customer payment terms.

The age distribution of gross trade receivable is as follows:	2015	2014
Not overdue	6,234	5,329
0–60 days overdue	449	577
61–120 days overdue	13	23
121–180 days overdue	21	9
More than 180 days overdue	<u>53</u>	<u>32</u>
Total gross receivables	6,770	5,970
Less allowance for doubtful receivables	<u>(360)</u>	<u>(79)</u>
Net trade receivables	<u>6,410</u>	<u>5,891</u>
Movement of allowance for bad debt:	2015	2014
Balance as at January 1	(79)	(48)
Realized losses for the year*	10	13
Charged to Income Statement as expense	<u>(291)</u>	<u>(44)</u>
Balance as at December 31	<u>(360)</u>	<u>(79)</u>

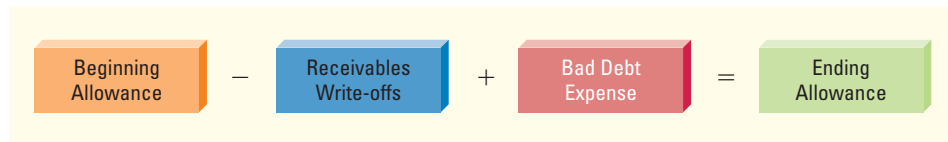
*for simplicity, including exchange rates translations and adjustments

estimates that 360 million kr is doubtful. Thus, the amount that LEGO expects to collect is DKK 6,410 million. This is called the net receivables because it is the amount of cash LEGO expects to realize in cash receipts eventually. This is also the number reported on the face of LEGO's Balance Sheet (Line 4 on Balance Sheet at the start of this chapter). We could say that as a percentage of total receivables, LEGO provided 5.32% allowance for doubtful receivables. But remember, LEGO does not predetermine this percentage and then apply it to total receivables to get the allowance.

Now, let's look at the last portion of LEGO's excerpt. It shows the movements of the allowance account: the beginning balance (which is equal to last year's ending balance), the amount used, the desired ending balance, and the necessary allowance for the year.

To recap, allowance was 79 million kr at the start of the year, and 10 million kr was used for receivable write-offs (more on this later). The total allowance after write-offs was 69 million kr (79 – 10), but a total of 360 is required based on LEGO's assessment of its customers' credit risks. Thus, the necessary allowance "top up" (and thus bad debt expense for the year) is 291 kr, calculated as 360 – 69 million kr.

You can think of the following formula to help you better understand the relationship between uncollectible-account expense, write-offs of receivables, and allowance for doubtful receivables accounts.



Suppose it is December 31, 2015, and LEGO's receivables show the following before the year-end adjustment (amounts in millions). The allowance account has a balance of 69 kr after receivable write-offs during the year.

Accounts Receivable	Allowance for Doubtful Receivables
6,770	10 79

These accounts are not yet ready for the financial statements because the allowance balance has not been adjusted to reflect LEGO's year-end assessment of the collectability of its receivables. You saw from earlier that the required ending balance of the allowance for doubtful receivables was 360 kr. To update the allowance, LEGO would make this adjusting entry at year-end:

A1			
	A	B	C
1	2015		
2	Dec. 31	Uncollectible-Account Expense	291
3		Allowance for Doubtful Receivables	291
4		<i>Recorded bad debt expense for the year.</i>	
5			

The expense decreases LEGO's assets and net income, as shown by the accounting equation.

Assets	=	Liabilities	+	Shareholders' Equity
-291	=	0	+	-291 (uncollectible account expense)

Now the Balance Sheet can report the amount that LEGO expects to collect from customers: 6,410 kr (6,770 kr less allowance of 360 kr).

Accounts Receivable	Allowance for Doubtful Receivables	Uncollectible-Account Expense
6,770	Write-off 10 Beg. Bal. 79	291
	Adj. 291	
	End. Bal. 360	
Net receivable, \$6,410		

Writing Off Uncollectible Accounts. Let's use a more specific example with customer balances to illustrate how businesses handle uncollectible receivables. Let's assume that at the beginning of 20X7 a Singapore division of LEGO had these accounts receivable (amounts in thousands):

Accounts Receivable— Fiesta	Accounts Receivable— Stop-N-Shop	Allowance for Doubtful Receivables
9	3	20
Accounts Receivable— Others		
88		
Total Accounts Receivable = \$100		Allowance = \$20
Accounts Receivable, Net = \$80		

Suppose that early in 20X7, LEGO's credit department determines that LEGO cannot collect from customers of Toys Kingdom and Hamleys. LEGO then writes off the receivables from these customers with the following entry:

A1				
	A	B	C	D
1	20X7			
2	Jan. 31	Allowance for Doubtful Receivables	12	
3		Accounts Receivable—Toys Kingdom		9
4		Accounts Receivable—Hamelys		3
5		Wrote off uncollectible receivables.		
6				

After the write-off, LEGO's accounts show these amounts:

Accounts Receivable— Toys Kingdom	Accounts Receivable— Hamelys	Allowance for Doubtful Receivables
9	3	12
		20
		8
Accounts Receivable— Others		
88		
Total Accounts Receivable = \$88		Allowance = \$8
Accounts Receivable, Net = \$80		

The receivables are reduced because it is now clear that these amounts will not be collected, and the allowance previously set aside is now “used” by the write-off.

Assets	=	Liabilities	+	Shareholders' Equity
+ 12	=	0	+	0
− 12				

The accounting equation shows that the write-off of uncollectibles has no effect on LEGO's total assets, no effect on current assets, and no effect on net accounts receivable. Notice that Accounts Receivable, Net is still \$80. There is no effect on net income either. Why is there no effect on net income? Net income is unaffected because the write-off of uncollectibles affects no expense account. It is merely a realization of the allowance of uncollectible receivables that has been provided in previous accounting periods. Naturally, at the year-end, when LEGO makes an adjustment for the required amount of allowance of uncollectible receivables, it may recognize more expenses as some of the allowance has been used by the earlier write-offs for Toys Kingdom and Hamleys.

Adjusting Ending Allowance for Doubtful Receivables. Let's continue with this example. Suppose by the end of 20X7, LEGO has receivables totaling \$200, and the balance of allowance for uncollectible receivables was the original \$20 less the write-offs of \$12 = \$8. The division also estimated that an appropriate level of allowance for the \$200 ending receivables is \$30. It will then adjust the allowance for uncollectible receivables to the desired ending balance by recording the following, and at the same time recognize an uncollectible receivables expense:

A1				
	A	B	C	D
1	20X7			
2	Dec. 31	Uncollectible Receivables Expense	22	
3		Allowance for Doubtful Receivables		
4		(\$30 – \$8)		22
5		<i>Recorded bad debt expense for the year.</i>		
6				

Recovery of Previously Written-off Receivables. Sometimes, accounts that have been written off may be partially recovered. This may take place after liquidation of the customer's business. Such recovery can be treated in one of two ways: (1) reverse the write-off entry or (2) decrease bad debt expense by the amount recovered. Both methods will eventually result in the same bad debt expense at the end of the period when new allowance (and expense) is calculated.

» Stop & Think

Suppose Mustafa Confectionery had beginning allowance for uncollectible account receivables of SAR (Saudi Arabia Riyal) 600,000. During the year, it wrote off SAR 200,000 of receivables. At the end of the year, an assessment of its receivables shows that credit quality has improved, and accordingly, the required ending balance for allowance for uncollectible account is SAR 300,000. What is the bad debt expense for the year?

Answer:

The current balance of the allowance for uncollectible account receivables account after receivable write-offs is SAR 400,000. Since the required ending allowance for uncollectible account receivables is SAR 300,000, i.e., lower than the current allowance, Mustafa will record a "negative" expense of SAR 100,000. This negative expense will result in an increase in profit for the year.

Direct Write-Off Method

There is another, less preferable, way to account for uncollectible receivables. Under the **direct write-off method**, the company waits until a specific customer's receivable proves uncollectible. Then the accountant writes off the customer's account and records Uncollectible-Account Expense, as follows (using assumed data):

A1			
	A	B	C
1	20X7		
2	Jan. 31	Uncollectible-Account Expense	12
3		Accounts Receivable—Fiesta	9
4		Accounts Receivable—Stop-N-Shop	3
5		Wrote off bad debts by direct write-off method.	
6			

The direct write-off method is not considered an accepted accounting practice for financial statement purposes. It is considered defective because it fails to take into account the possibility of impairment of the receivables at Balance Sheet date. As a result, receivables are always reported at their full amount, which is more than the business expects to collect. *Assets on the Balance Sheet may be overstated.* Because of these deficiencies, LEGO and virtually all companies use the allowance method for preparing their financial statements.

Sometimes you may see the concept of the direct write-off method being used for income tax purposes. It is one of several sources of temporary differences that may arise between net income for financial reporting purposes and net income for income tax purposes. We will discuss other differences between book and taxable income in later chapters.

Computing Cash Collections from Customers

A company earns revenue and then collects the cash from customers. For LEGO and most other companies, there is a time lag between earning the revenue and collecting the cash. Collections from customers are the single most important source of cash for any business. You can compute a company's collections from customers by analyzing its Accounts Receivable account. Receivables typically hold only three types of transactions, as reflected in the five elements of the following Accounts Receivable account balance (amounts assumed):

Accounts Receivable			
Beg. Bal. (left over from last period)	200	Write-offs of uncollectibles	100 ^b
Sales (or service) revenue	1,800 ^a	Collections from customers	X = 1,500 ^c
End. Bal. (carries over to next period)	400		

^aThe journal entry that places revenue into the accounts receivable account is

A1			
	A	B	C
1	Accounts Receivable	1,800	
2	Sales (or Service) Revenue		1,800
3			

^bThe journal entry for write-offs is

A1			
	A	B	C
1	Allowance for Uncollectibles	100	
2	Accounts Receivable		100
3			

^c The journal entry that places collections into the receivable account is

A1			
	A	B	C
1	Cash	1,500	
2	Accounts Receivable		1,500
3			

CASH
FLOW



Suppose you know all these amounts *except* collections from customers. You can compute collections by solving for x in the T-account. The following diagram may help you link the numbers. Collections from customers, x , is therefore \$1,500.



Often write-offs are unknown and may be omitted. Then the computation of collections becomes an approximation.

Accounting for Notes Receivable

As stated earlier, notes receivable are more formal than accounts receivable. Notes receivable due within one year or less are current assets. Notes due beyond one year are *long-term receivables* and are reported as long-term assets. Some notes receivable are collected in installments. The portion due within one year is a current asset and the remainder is long term. LEGO may hold a \$20,000 note receivable from a customer, but only the \$6,000 the customer must pay within one year is a current asset of LEGO.

Before launching into the accounting for notes receivable, let's define some key terms:

Creditor. The party to whom money is owed. The **creditor** is also called the *lender*.

Debtor. The party that borrowed and owes money on the note. The **debtor** is also called the *maker* of the note or the *borrower*.

Interest. Interest is the cost of borrowing money. The interest is stated as an annual percentage rate.

Maturity date. The date on which the debtor must pay the note.

Maturity value. The sum of principal and interest on the note payable on maturity.

Principal. The amount of money borrowed by the debtor.

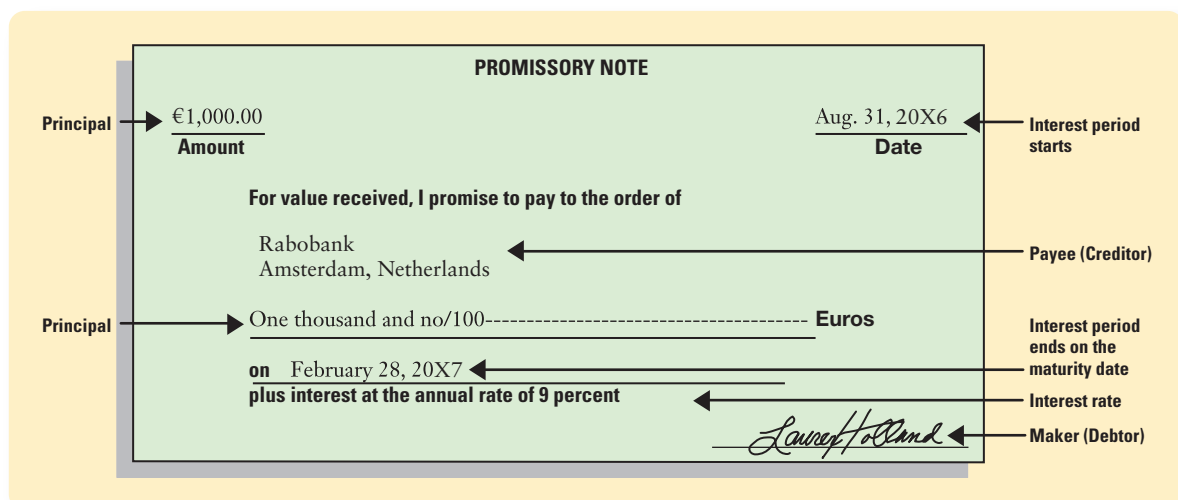
Term. The length of time from when the note was signed by the debtor to when the debtor must pay the note.

There are two parties to a note:

- The *creditor* has a note receivable.
- The *debtor* has a note payable.

Exhibit 5-16 is a typical promissory note.

Exhibit 5-16 | A Promissory Note



The **principal** amount of the note (€1,000) is the amount borrowed by the debtor, lent by the creditor. This six-month note receivable runs from August 31, 20X6, to February 28, 20X7, when Lauren Holland (the maker) promises to pay Rabobank (the creditor) the principal of €1,000 plus 9% annual interest. Interest is revenue to the creditor (Rabobank, in this case).

Consider the promissory note in Exhibit 5-16. After Lauren Holland signs the note, Rabobank gives her €1,000 cash. The bank's entries follow, assuming a December 31 year-end for Rabobank:

A1				
	A	B	C	D
1	20X6			
2	Aug. 31	Note Receivable—L. Holland	1,000	
3		Cash		1,000
4		Made a loan.		
5				

Note Receivable— L. Holland	
1,000	

The bank gave one asset, cash, in return for another asset, a note receivable, so total assets did not change.

Rabobank earns interest revenue during September, October, November, and December. At December 31, the bank accrues 9% interest revenue for four months as follows:

A1				
	A	B	C	D
1	20X6			
2	Dec. 31	Interest Receivable ($€1,000 \times 0.09 \times 4/12$)	30	
3		Interest Revenue		30
4		Accrued interest revenue.		
5				

The bank's assets and revenues increase. Rabobank reports these amounts in its financial statements at December 31, 20X7:

Balance Sheet	
Current assets:	
Note receivable	€1,000
Interest receivable.....	30
Income Statement	
Interest revenue.....	€ 30

The bank collects the note on February 28, 20X7, and records:

A1				
	A	B	C	D
1	20X7			
2	Feb. 28	Cash	1,045	
3		Note Receivable—L. Holland		1,000
4		Interest Receivable		30
5		Interest Revenue ($€1,000 \times 0.09 \times 2/12$)		15
6		Collected note at maturity.		
7				

This entry zeroes out Note Receivable and Interest Receivable and also records the interest revenue earned in 20X7.

Note Receivable— L. Holland	
1,000	1,000

In its 20X7 financial statements, the only item that Rabobank will report is the interest revenue of €15 that was earned in 20X7. There's no note receivable or interest receivable on the Balance Sheet because those items were zeroed out when the bank collected the note at maturity.

Three aspects of the interest computation deserve further mention:

1. Interest rates are always for an annual period, unless stated otherwise. In this example, the annual interest rate is 9%. At December 31, 20X6, Rabobank accrues interest revenue for four months. The interest computation is

Principal	×	Interest Rate	×	Time	=	Amount of Interest
€1,000	×	0.09	×	4/12	=	€30

2. The time element (4/12) is the fraction of the year that the note has been in force during 20X6.
3. Interest is often computed for a number of days. For example, suppose you loaned out €10,000 on April 10. The note receivable runs for 90 days and specifies interest at 8%.
 - a. Interest starts accruing on April 11 and runs for 90 days, ending on the due date, July 9, as follows:

Month	Number of Days That Interest Accrues
April	20
May	31
June	30
July	9
Total	90

- b. The interest computation is $\$10,000 \times 0.08 \times 90/365 = \197

Some companies sell goods and services on notes receivable (versus selling on accounts receivable). This often occurs when the payment term extends beyond the customary accounts receivable period of 30 to 60 days.

Suppose that on March 20, 20X7, LEGO sells a large amount of merchandise to Robinson Department Store. LEGO gets Robinson's three-month promissory note plus 10% annual interest. On initial recognition, Nestlé would debit Notes Receivable and credit Sales Revenue.

A company may also accept a note receivable from a trade customer whose account receivable is past due. The company then debits Notes Receivable and credits Accounts Receivable. We would say the company "received a note receivable on account." Now let's examine some strategies to speed up cash flow.

CASH FLOW



How to Speed Up Cash Flow

All companies want speedy cash receipts. Rapid cash flow finances new products, research, and development. Thus, companies such as LEGO find ways to collect cash quickly. Two common strategies generate cash quickly.

Credit Card or Bankcard Sales. The merchant sells merchandise and lets the customer pay with credit cards such as American Express, VISA, or MasterCard. This strategy may increase

sales, but the added revenue comes at a cost, which is typically about 2% to 3% of the total amount of the sale. Let's see how credit cards and bankcards work from the seller's perspective.

Suppose Fujitsu sells computers for \$2,000, and the customer pays with a VISA card. Fujitsu records the sale as follows:

A1			
	A	B	C
1	Cash	1,960	
2	Credit Card Processing Fee	40	
3	Sales Revenue		2,000
4	Recorded bankcard sales.		
5			

Assets	=	Liabilities	+	Shareholders' Equity	
+ 1,960	=	0	+	+ 2,000	(sales revenue)
				- 40	(credit card fees)

Fujitsu enters the transaction in the credit card machine. The machine, linked to a VISA server, automatically credits Fujitsu's account for a discounted portion, say \$1,960, of the \$2,000 sale amount. Two percent (\$40) goes to VISA. To Fujitsu, the credit card processing fee is an operating expense that reduces Fujitsu's net profit for the year.

Selling (Factoring) Receivables. LEGO makes some large sales to toy stores on account, debiting Accounts Receivable and crediting Sales Revenue. LEGO might then sell these accounts receivable to another business, called a *factor*. The factor earns revenue by paying a discounted price for the receivable and then hopefully collecting the full amount from the customer. The benefit to LEGO is the immediate receipt of cash. The biggest disadvantage of factoring is that it is often quite expensive, when compared to the costs of retaining the receivable on the books and ultimately collecting the full amount. In addition, the company that factors its receivables may lose control over the collection process and yet be responsible for any bad debts that may arise after the factoring. For these reasons, factoring is often not used by companies who have other less costly means to raise cash, such as short-term borrowing from banks. Factoring may be used by start-up companies with insufficient credit history to obtain loans at a reasonable cost, by companies with weaker credit history, or by companies that are already saddled with a significant amount of debt.

To illustrate selling, or *factoring*, accounts receivable, suppose a company wishes to speed up cash flow and therefore sells \$100,000 of accounts receivables, receiving cash of \$95,000. The company would record the sale of the receivables as follows:

A1			
	A	B	C
1	Cash	95,000	
2	Financing Expense	5,000	
3	Accounting Receivable		100,000
4	Sold accounts receivable.		
5			

Again, Financing Expense is an operating expense, with the same effect as Interest Expense. Discounting a note receivable is similar to selling an account receivable. However, the credit is to Notes Receivable (instead of Accounts Receivable).

Notice the high price (5% of the face amount, or \$5,000) the company has had to sacrifice in order to collect the cash immediately, as opposed to waiting 30 to 60 days to collect the full amount. Therefore, if the company can afford to wait, it will probably not engage in factoring in order to collect the full amount of the receivables.

5 Evaluate a company's ability to collect receivables

EVALUATE A COMPANY'S ABILITY TO COLLECT RECEIVABLES

After a business makes a credit sale, the next step is collecting the receivable. **Receivable collection period**, also called the *days' sales in receivables*, or *days sales outstanding*, tells a company how long it takes to collect its average level of receivables. Shorter is better because cash is coming in more quickly. The longer the collection period, the less cash is available to pay bills and expand.

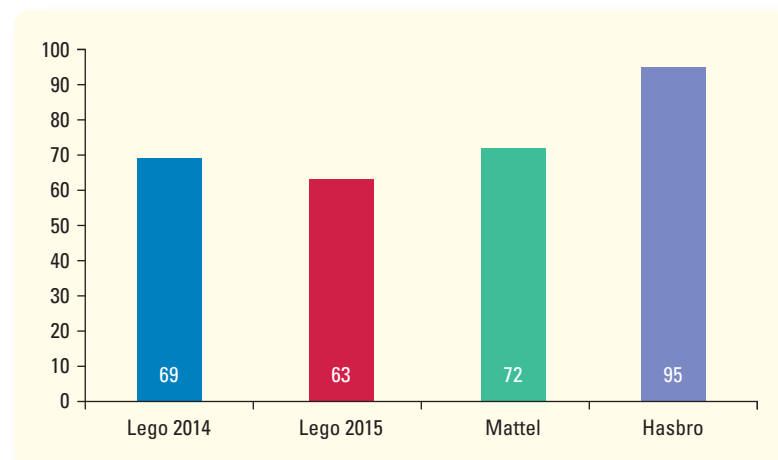
The receivable collection period can be computed in two logical steps. First, we can calculate the **receivables turnover**. The receivable turnover is the number of times average receivables are converted into cash in a year. This is calculated as Sales Revenue divided by Average Receivables. Second, we can convert this number of times into days by dividing 365 by the receivable turnover. Net sales come from the Income Statement, and the receivables amounts are taken from the Balance Sheet. Average receivables is the simple average of the beginning and ending balances.

Receivable turnover and resident period	LEGO 2015	LEGO 2014
$\frac{\text{Sales}}{\text{Average receivables}^*}$	$\frac{35,780}{6,150.5} = 5.82 \text{ times}$	$\frac{28,578}{5,380.5} = 5.31 \text{ times}$
365/Receivable turnover	365/5.82 = 63 days	365/5.31 = 69 days
<i>*Average = (beginning + ending balance)/2, which is (6,410 + 5,981)/2 for 2015</i>		

It takes LEGO an average of 63 days to collect its receivables in 2015 (compared to 69 days in 2014). It appears that LEGO is more effective in collecting its receivables in 2015. To evaluate LEGO's collection period of 69 days, we can also compare it to the credit terms that LEGO offers customers when the company makes a sale, as well as the number of days on average that creditors typically allow LEGO to pay them without penalty. Suppose LEGO makes sales on "net 30" terms, which means that customers should pay LEGO within 30 days of the sale. LEGO's collection period of 69 days is longer than the credit terms it gives to its customers. After all, some customers do drag out their payments. And, as we've seen, some customers don't pay at all. On the other hand, if LEGO's short-term creditors expect payment of their accounts payable within 30 days, LEGO may need to borrow cash in order to pay its creditors on time, if it doesn't have enough cash on hand.

Remember that it is a good practice to compare a company's performance to some other benchmarks or industry average. Let's compare LEGO's receivable collection period to other big names in the toy industry, as shown in Exhibit 5-17. Mattel and Hasbro have receivable collection periods of 72 days and 95 days, respectively. We can conclude that LEGO's receivable collection period is much better than Mattel and Hasbro.

Exhibit 5-17 | Receivable Collection Period Comparisons



Companies watch their collection periods closely. Whenever collections slow down, the business must find other sources of financing, such as borrowing or selling receivables. During recessions, customers may take an even longer time to pay, and a longer collection period may be unavoidable.

End-of-Chapter

Summary Problem

Superior Technical Resources' (STR's) Balance Sheet at December 31, 20X6, reported STR uses the aging method to account for uncollectible receivables.

	(In millions)
Accounts receivable.....	\$382
Allowance for doubtful accounts.....	(92)

Requirements

- How much of the December 31, 20X6, balance of accounts receivables did STR expect to collect? Stated differently, what was the net realizable value of STR's receivables?
- Journalize, without explanations, 20X7 entries for STR:
 - Write-offs of uncollectible accounts receivable totaling \$58 million. Prepare a T-account for Allowance for Doubtful Accounts and post to this account. Show its unadjusted balance at December 31, 20X7.
 - December 31, 20X7, aging of receivables, which indicates that \$47 million of the total receivables of \$409 million is uncollectible at year-end. Post to Allowance for Doubtful Accounts, and show its adjusted balance at December 31, 20X7.
- Show how STR's receivables and the related allowance will appear on the December 31, 20X7, Balance Sheet.
- Show how STR's Income Statement will report for the foregoing transactions.

Answers

Requirement 1

	(In millions)
Net realizable value of receivables (\$382 – \$92)	\$290

Requirement 2

A1			
A	B	C	D
1		(In millions)	
2	a. Allowance for Doubtful Accounts	58	
3	Accounts Receivable		58
4			

Allowance for Doubtful Accounts			
		Dec. 31, 20X6	92
20X7 Write-offs	58		
		Unadjusted balance at Dec. 31, 20X7	34

A1				
	A	B	C	D
1	c.	Doubtful-Account Expense (\$47 – \$34)	13	
2		Allowance for Doubtful Accounts		13
3				

Allowance for Doubtful Accounts	
	Dec. 31, 20X7 Unadj. bal. 34
	2011 Expense 13
	Dec. 31, 20X7 Adj. bal. 47

Requirement 3

(In millions)	
Accounts receivable.....	\$409
Allowance for doubtful accounts.....	(47)

Requirement 4

(In millions)	
Expenses: Doubtful-account expense for 20X7 (\$40 + \$13)	\$53

REVIEW | Internal Control, Cash, and Receivables

Quick Check (Answers are given at the end of the chapter.)

- Each of the following is an example of a control procedure, *except*
 - a sound marketing plan.
 - limited access to assets.
 - sound personnel procedures.
 - separation of duties.

Laurie Corporation has asked you to prepare its bank reconciliation at the end of the current month. Answer **questions 2–6** using the following code letters to indicate how the item described would be reported on the bank reconciliation.

- Add to the book balance
 - Deduct from the book balance
 - Add to the bank balance
 - Deduct from the bank balance
 - Does not belong on the bank reconciliation
- A check for \$925 written by Laurie during the current month was erroneously recorded as a \$952 payment.
 - A \$400 deposit made on the last day of the current month did not appear on this month's bank statement.

4. The bank statement showed interest earned of \$75.
5. The bank statement included a check from a customer that was marked NSF.
6. The bank statement showed the bank had credited Laurie's account for a \$750 deposit made by Lawrence Company.
7. Which of the following reconciling items does not require a journal entry?
 - a. NSF check
 - b. Deposit in transit
 - c. Bank service charge
 - d. Bank collection of note receivable
8. A cash budget helps control cash by
 - a. helping to determine whether additional cash is available for investments or new financing is needed.
 - b. ensuring accurate cash records.
 - c. developing a plan for increasing sales.
 - d. all of the above.
9. Accounts Receivable has a debit balance of \$2,400, and the Allowance for Uncollectible Accounts has a credit balance of \$400. A \$90 account receivable is written off. What is the amount of net receivables (net realizable value) after the write-off?
 - a. \$1,910
 - b. \$2,000
 - c. \$2,090
 - d. \$2,310
10. Aurora Company received a four-month, 6% per annum, \$2,800 note receivable on December 1. The adjusting entry on December 31 will
 - a. debit Interest Receivable \$14.
 - b. credit Interest Revenue \$14.
 - c. both a and b.
 - d. credit Interest Revenue \$168.
11. If the adjusting entry to accrue interest on a note receivable is omitted, then
 - a. assets, net income, and shareholders' equity are overstated.
 - b. assets, net income, and shareholders' equity are understated.
 - c. assets are overstated, net income is understated, and shareholders' equity is understated.
 - d. liabilities are understated, net income is overstated, and shareholders' equity is overstated.
12. Net sales total \$803,000. Beginning and ending accounts receivable are \$80,000 and \$74,000, respectively. Calculate receivables collection period.
 - a. 10 days
 - b. 35 days
 - c. 34 days
 - d. 36 days

Accounting Vocabulary

audit (p. 263) A periodic examination of a company's financial statements and the accounting systems, controls, and records that produce them. Audits may be either external or internal. External audits are usually performed by certified public accountants (CPAs).

bank collection (p. 273) Collection of money by the bank on behalf of a depositor.

bank reconciliation (p. 273) A document explaining the reasons for the difference between a depositor's records and the bank's records about the depositor's cash.

bank statement (p. 271) Document showing the beginning and ending balances of a particular bank account listing the month's transactions that affected the account.

budget (p. 263) A quantitative expression of a plan that helps managers coordinate the entity's activities.

cash budget (p. 263) A budget that projects the entity's future cash receipts and cash disbursements.

cash equivalent (p. 278) Investments such as time deposits, certificates of deposit, or high-grade government securities that are considered so similar to cash that they are combined with cash for financial disclosure purposes on the Balance Sheet and cash flow statement.

check (p. 270) Document instructing a bank to pay the designated person or business the specified amount of money.

computer virus (p. 265) A malicious program that enters a company's computer system by e-mail or other means and destroys program and data files.

controller (p. 263) The chief accounting officer of a business.

creditor (p. 290) The party to whom money is owed; also called the lender.

debtor (p. 290) The party that borrowed and owes money on the note; also called the maker of the note or the borrower.

deposits in transit (p. 273) A deposit recorded by the company but not yet by its bank.

direct write-off method (p. 288) A method of accounting for bad debts.

doubtful-account expense (p. 282) Another name for *uncollectible-account expense*.

electronic fund transfer (EFT) (p. 272) System that transfers cash by electronic communication rather than by paper documents.

exception reporting (p. 263) Identifying data that is not within “normal limits” so that managers can follow up and take corrective action. Exception reporting is used in operating and cash budgets to keep company profits and cash flow in line with management’s plans.

fraud (p. 255) An intentional misrepresentation of facts, made for the purpose of persuading another party to act in a way that causes injury or damage to that party.

fraud triangle (p. 258) The three elements that are present in almost all cases of fraud. These elements are pressure, opportunity, and rationalization on the part of the perpetrator.

fraudulent financial reporting (p. 257) Fraud perpetrated by management by preparing misleading financial statements.

imprest system (p. 269) A way to account for petty cash by maintaining a constant balance in the petty cash account, supported by the fund (cash plus payment tickets) totaling the same amount.

interest (p. 290) The borrower’s cost of renting money from a lender. Interest is revenue for the lender and expense for the borrower.

internal control (p. 259) Organizational plan and related measures adopted by an entity to safeguard assets, encourage adherence to company policies, promote operational efficiency, and ensure accurate and reliable accounting records.

lapping (p. 262) A fraudulent scheme to steal cash through misappropriating certain customer payments and posting payments from other customers to the affected accounts to cover it up. Lapping is caused by weak internal controls (i.e., not segregating the duties of cash handling and accounts receivable bookkeeping, allowing the bookkeeper improper access to cash, and not appropriately monitoring the activities of those who handle cash).

lock-box system (p. 263) A system of handling cash receipts by mail whereby customers remit payment directly to the bank, rather than through the entity’s mail system.

maturity date (p. 290) The date on which the debtor must pay the note.

maturity value (p. 290) The sum of principal and interest on the note.

misappropriation of assets (p. 257) Fraud committed by employees by stealing assets from the company.

non-sufficient funds (NSF) cheque (p. 273) A “hot” cheque, one for which the payer’s bank account has insufficient money to pay the cheque. NSF cheques are cash receipts that turn out to be worthless.

operating budget (p. 263) A budget of future net income. The operating budget projects a company’s future revenue and expenses. It is usually prepared by line item of the company’s Income Statement.

outstanding cheques (p. 273) A cheque issued by the company and recorded on its books but not yet paid by its bank.

petty cash (p. 269) Fund containing a small amount of cash that is used to pay minor amounts.

phishing (p. 265) Creating bogus websites for the purpose of stealing unauthorized data, such as names, addresses, social security numbers, bank account, and credit card numbers.

principal (p. 291) The amount borrowed by a debtor and lent by a creditor.

receivable collection period (p. 294) the average number of days to collect receivables from customers, calculated by dividing 365 by receivable turnover. Also called *days’ sales in receivables* or *days sales outstanding*.

receivable turnover (p. 294) the number of times receivables are collected during the year, approximated by taking total sales divided by average receivables. Used in calculating *receivable collection period*.

receivables (p. 280) Monetary claims against a business or an individual, acquired mainly by selling goods or services and by lending money.

remittance advice (p. 270) An optional attachment to a cheque (sometimes a perforated tear-off document and sometimes capable of being electronically scanned) that indicates the payer, date, and purpose of the cash payment. The remittance advice is often used as the source documents for posting cash receipts or payments.

term (p. 290) The length of time from when the note was signed by the debtor to when the debtor must pay the note.

treasurer (p. 262) In a large company, the department that has total responsibility for cash handling and cash management. This includes cash budgeting, cash collections, writing cheques, investing excess funds, and making proposals for raising additional cash when needed.

Trojan horse (p. 265) A malicious program that hides within legitimate programs and acts like a computer virus.

uncollectible-account expense (p. 282) Cost to the seller of extending credit. Arises from the failure to collect from credit customers. Also called doubtful-account expense, or bad-debt expense.

ASSESS YOUR PROGRESS

Short Exercises

S5-1. (Learning Objective 1: Defining fraud) Define “fraud.” List and briefly discuss the three major components of the fraud triangle.

L0 1

■ writing assignment

S5-2. (Learning Objective 1: Explaining and describing characteristics of an effective system of internal control) Explain why separation of duties is often described as the cornerstone of internal control for safeguarding assets. Describe what can happen if the same person has custody of an asset and also accounts for the asset.

L0 1

S5-3. (Learning Objective 1: Explaining the role of internal control) Cash may be a small item on the financial statements. Nevertheless, internal control over cash is very important. Why is this true?

L0 1

S5-4. (Learning Objective 2: Applying internal controls over cash receipts) Candice Cassidy sells memberships to the Phoenix Symphony Association in Vienna, Austria. The Symphony’s procedure requires Cassidy to write a patron receipt for all memberships sold. The receipt forms are prenumbered. Cassidy is having personal financial problems and she stole \$600 received from a customer. To hide her theft, Cassidy destroyed the company copy of the receipt that she gave the patron. What will alert manager Stephanie Stevens that something is wrong?

L0 2

S5-5. (Learning Objective 2: Applying internal control over cash payments by check) Answer the following questions about internal control over cash payments:

L0 2

1. Payment by check carries three controls over cash. What are they?
2. Suppose a purchasing agent receives the goods that he purchases and also approves payment for the goods. How could a dishonest purchasing agent cheat his company? How do companies avoid this internal control weakness?

S5-6. (Learning Objective 3: Preparing a bank reconciliation) The Cash account of Randell Corp. reported a balance of \$2,500 at October 31. Included were outstanding checks totaling \$600 and an October 31 deposit of \$350 that did not appear on the bank statement. The bank statement, which came from Park Bank, listed an October 31 balance of \$3,280. Included in the bank balance was an October 30 collection of \$560 on account from a customer who pays the bank directly. The bank statement also shows a \$20 service charge, \$30 of interest revenue that Randell earned on its bank balance, and an NSF check for \$40.

L0 3

Prepare a bank reconciliation to determine how much cash Randell actually has at October 31.

S5-7. (Learning Objective 3: Recording transactions from a bank reconciliation) After preparing Randell Corp.’s bank reconciliation in Short Exercise 5-6, make the company’s journal entries for transactions that arise from the bank reconciliation. Date each transaction October 31, and include an explanation with each entry.

L0 3

■ writing assignment

S5-8. (Learning Objective 3: Using a bank reconciliation as a control device) Jessica Smith manages Jones Advertising. Smith fears that a trusted employee has been stealing from the company. This employee receives cash from clients and also prepares the monthly bank reconciliation. To check up on the employee, Smith prepares her own bank reconciliation, as follows:

L0 3

A1				
	A	B	C	D
1	Jones Advertising Bank Reconciliation October 31, 20X7			
2	Bank		Books	
3	Balance, October 31	\$ 4,600	Balance, October 31	\$ 3,920
4	Add:		Add:	
5	Deposit in transit	800	Bank collections	1,100
6			Interest revenue	20
7	Less:		Less:	
8	Outstanding checks	(900)	Service charge	(40)
9	Adjusted bank balance	\$ 4,500	Adjusted book balance	\$ 5,000
10				

Does it appear that the employee has stolen from the company? If so, how much? Explain your answer. Which side of the bank reconciliation shows the company's true cash balance?

L0 3

S5-9. (Learning Objective 3: Preparing a cash budget) Crescent Artichoke Growers (CAG) is a major food cooperative. Suppose CAG begins 20X6 with cash of \$14 million. CAG estimates cash receipts during 20X6 will total \$124 million. Planned payments will total \$106 million. To meet daily cash needs next year, CAG must maintain a cash balance of at least \$22 million. Prepare the organization's cash budget for 20X6.

L0 4

S5-10. (Learning Objective 4: Applying the allowance method to account for uncollectibles) Use the information from the following journal entries of Green Leaves Furniture Restoration to answer the questions below:

1	A	B	C	D
1		Journal Entry		
2		Accounts	Debit	Credit
3	1.	Accounts Receivable	1,200,000	
4		Sales Revenue		1,200,000
5				
6	2.	Cash	980,000	
7		Accounts Receivable		980,000
8				
9	3.	Allowance for Uncollectible Accounts	18,000	
10		Accounts Receivable		18,000
11				
12	4.	Uncollectible-Account Expense	40,000	
13		Allowance for Uncollectible Accounts		40,000
14				

Requirements

1. Start with Accounts Receivable's beginning balance (\$48,000) and then post to the Accounts Receivable T-account. How much do Green Leaves Furniture Restoration's customers owe the company at December 31?
2. Start with the Allowance account's beginning credit balance (\$10,480) and then post to the Allowance for Uncollectible Accounts T-account. How much of the receivables at December 31 does Green Leaves Furniture Restoration expect *not* to collect?
3. At December 31, how much cash does Green Leaves Furniture Restoration expect to collect on its accounts receivable?

L0 4

S5-11. (Learning Objective 4: Applying the allowance method to account for uncollectibles) Goode and Devlin, a law firm, started 20X6 with accounts receivable of \$31,000 and an allowance for uncollectible accounts of \$4,200. The 20X6 service revenues on account totaled \$157,000, and cash collections on account totaled \$121,000. During 20X6, Goode and Devlin wrote off uncollectible accounts receivable of \$3,200. At December 31, 20X6, the aging of accounts receivable indicated that Goode and Devlin will not collect \$1,850 of its accounts receivable.

Journalize Goode and Devlin's (a) service revenue, (b) cash collections on account, (c) write-offs of uncollectible receivables, and (d) uncollectible-account expense for the year. Explanations are not required. Prepare a T-account for Allowance for Uncollectible Accounts to show your computation of uncollectible-account expense for the year.

L0 4

S5-12. (Learning Objective 4: Applying the allowance method to account for uncollectibles) Perform the following accounting for the receivables of Robbins and Williams, a law firm, at December 31, 20X6.

Requirements

- Start with the beginning balances for these T-accounts:
 - Accounts Receivable, \$89,000
 - Allowance for Uncollectible Accounts, \$6,000
 Post the following 20X6 transactions to the T-accounts:
 - Service revenue of \$753,000, all on account
 - Collections on account, \$774,000
 - Write-offs of uncollectible accounts, \$8,000
 - Uncollectible-account expense (allowance method), \$12,000
- What are the ending balances of Accounts Receivable and Allowance for Uncollectible Accounts?
- Show how Robbins and Williams will report accounts receivable on its Balance Sheet at December 31, 20X6.

S5-13. (Learning Objectives 4: Answering practical questions about receivables) Answer these questions about receivables and uncollectibles. For the true-false questions, explain any answers that turn out to be false.

L0 4

- True or false? The direct write-off method of accounting for uncollectibles understates assets.
- California Bank lent \$240,000 to Sacramento Company on a six-month, 8% note. Which party has interest receivable? Which party has interest payable? Interest expense? Interest revenue? How much interest will these organizations record one month after Sacramento Company signs the note?
- When California Bank accrues interest on the Sacramento Company note, show the directional effects on the bank's assets, liabilities, and equity (increase, decrease, or no effect).
- True or false? Credit sales increase receivables. Collections and write-offs decrease receivables.
- Which receivables figure (the *total* amount that customers *owe* the company, or the *net* amount the company expects to collect) is more interesting to investors as they consider buying the company's shares? Give your reason.
- Show how to determine net accounts receivable.

S5-14. (Learning Objective 4: Accruing interest receivable and collecting a note receivable)

L0 4

On August 31, 20X6, Nancy Thompson borrowed \$2,200 from Yellow Interstate Bank. Thompson signed a note payable, promising to pay the bank principal plus interest on August 31, 20X7. The interest rate on the note is 9%. The accounting year of Yellow Interstate Bank ends on June 30, 20X7. Journalize Yellow Interstate Bank's (a) lending money on the note receivable at August 31, 20X6, (b) accrual of interest at June 30, 20X7, and (c) collection of principal and interest at August 31, 20X7, the maturity date of the note.

S5-15. (Learning Objective 5: Evaluating the current ratio and days' sales in receivables)

L0 5

West Highland Clothiers reported the following amounts in its 20X7 financial statements. The 20X6 amounts are given for comparison.

	20X7	20X6
Current assets:		
Cash.....	\$ 10,200	\$ 9,700
Short-term investments.....	19,000	14,000
Accounts receivable.....	\$86,000	\$77,000
Less: Allowance for uncollectibles.....	(6,400)	(6,100)
Inventory.....	79,600	70,900
Prepaid insurance.....	2,400	2,300
Total current assets.....	300,200	287,400
Total current liabilities.....	103,000	111,000
Net sales.....	852,000	736,000

Requirements

1. Compute West Highland's current ratio at the end of 20X7. Round to two decimal places.
2. Compute West Highland's receivables collection period for 20X7.

Exercises MyLab Accounting

Select Group A and Group B exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

LO 1 2

■ writing assignment

E5-16A. (Learning Objectives 1, 2: Learning about fraud; identifying internal control weaknesses) Identify the internal control weakness in the following situations. State how the person can hurt the company.

- a. Jim Morris works as a security guard at SAFETY parking in Budapest. Morris has a master key to the cash box where customers pay for parking. Each night Morris prepares the cash report that shows (a) the number of cars that parked on the lot and (b) the day's cash receipts. Lucy Carrols, the SAFETY treasurer, checks Morris's figures by multiplying the number of cars by the parking fee per car. Carrington then deposits the cash in the bank.
- b. Cecilia Albom is the purchasing agent for Marshfield Golf Equipment. Albom prepares purchase orders based on requests from division managers of the company. Albom faxes the purchase order to suppliers who then ship the goods to Marshfield. Albom receives each incoming shipment and checks it for agreement with the purchase order and the related invoice. She then routes the goods to the respective division managers and sends the receiving report and the invoice to the accounting department for payment.

LO 2

E5-17A. (Learning Objective 2: Identifying internal control strengths and weaknesses) The following situations describe two cash payment situations and two cash receipt situations. In each pair, one set of internal controls is better than the other. Evaluate the internal controls in each situation as strong or weak, and give the reason for your answer.

Cash payments:

- a. Stone & Grain, Inc., policy calls for project supervisors to purchase the equipment needed for jobs. The supervisors then submit the paid receipts to the home office for reimbursement. This policy enables supervisors to get the equipment quickly and keep construction jobs moving.
- b. Tony McGraw Construction policy calls for construction supervisors to request the equipment needed for their jobs. The home office then purchases the equipment and has it shipped to the construction site.

Cash receipts:

- a. Cash received by mail at Wright Orthopedic Clinic goes to the mail room, where a mail clerk opens envelopes and totals the cash receipts for the day. The mail clerk forwards customer checks to the cashier for deposit in the bank and forwards the remittance advices to the accounting department for posting credits to customer accounts.
- b. At Cristy Auto Parts, cash received by mail goes straight to the accountant, who debits Cash and credits Accounts Receivable to record the collections from customers. The Cristy accountant then deposits the cash in the bank.

LO 1 2

■ writing assignment

E5-18A. (Learning Objectives 1, 2: Learning about fraud; correcting an internal control weakness) Barry Floyd served as executive director of Downtown Kalamazoo, an organization created to revitalize Kalamazoo. Over the course of 13 years, Floyd embezzled \$348,000. How did Floyd do it? By depositing subscriber cash receipts in his own bank account, writing Downtown Kalamazoo checks to himself, and creating phony entities to which Downtown Kalamazoo wrote checks.

Downtown Kalamazoo was led by a board of directors comprised of civic leaders. Floyd's embezzlement went undetected until Downtown Kalamazoo couldn't pay its bills. Give four ways in which Floyd's embezzlement could have been prevented.

LO 2

E5-19A. (Learning Objective 2: Evaluating internal control over cash receipts) McKinley stores use point-of-sale terminals as cash registers. The register shows the amount of each sale, the

cash received from the customer, and any change returned to the customer. The machine also produces a customer receipt but keeps no record of transactions. At the end of the day, the clerk counts the cash in the register and gives it to the cashier for deposit in the company's bank account.

Write a memo to convince the store manager that there is an internal control weakness over cash receipts. Identify the weakness that gives an employee the best opportunity to steal cash, and state how to prevent such a theft.

E5-20A. (Learning Objective 2: Evaluating internal control over cash payments) New Pastures Golf Company manufactures a popular line of golf clubs. New Pastures Golf employs 176 workers and keeps their employment records on time sheets that show how many hours the employee works each week. On Friday the shop foreman collects the time sheets, checks them for accuracy, and delivers them to the payroll department for preparation of paychecks. The treasurer signs the paychecks and returns the checks to the payroll department for distribution to the employees.

L0 2

Identify the main internal control weakness in this situation, state how the weakness can hurt New Pastures Golf, and propose a way to correct the weakness.

E5-21A. (Learning Objective 3: Classifying bank reconciliation items) The following items appear on a bank reconciliation:

L0 3

1. ☐ Service charge
2. ☐ Deposits in transit
3. ☐ Outstanding checks
4. ☐ Bank error: The bank credited our account for a deposit made by another bank customer.
5. ☐ Book error: We debited Cash for \$200. The correct debit was \$2,000.
6. ☐ NSF check
7. ☐ Bank collection of a note receivable on our behalf

Classify each item as (a) an addition to the bank balance, (b) a subtraction from the bank balance, (c) an addition to the book balance, or (d) a subtraction from the book balance.

E5-22A. (Learning Objective 3: Preparing a bank reconciliation) D.J. Holt's checkbook lists the following:

L0 3

Date	Check No.	Item	Check	Deposit	Balance
8/1					\$ 675
4	852	Art Cafe	\$ 40		635
9		Dividends received		\$ 120	755
13	853	General Tire Co.	45		710
14	854	QuickMobil	58		652
18	855	Cash	65		587
26	856	Woodway Baptist Church	95		492
28	857	Bent Tree Apartments	275		217
31		Paycheck		1,010	1,227

The August bank statement shows

Balance		\$675
Add: Deposits		120
Debit checks:	No. Amount	
	852 \$40	
	853 45	
	854 85*	
	855 65	(235)
Other charges:		
NSF check	\$20	
Service charge	10	(30)
Balance		<u>\$530</u>

*This is the correct amount for check number 854.

Requirement

1. Prepare Holt's bank reconciliation at August 31, 20X7.

L0 3

E5-23A. (Learning Objective 3: Preparing a bank reconciliation) Evan Stephens operates a bowling alley. He has just received the monthly bank statement at April 30 from City National Bank, and the statement shows an ending balance of \$585. Listed on the statement are an EFT rent collection of \$330, a service charge of \$8, two NSF checks totaling \$135, and a \$10 charge for printed checks. In reviewing his cash records, Stephens identifies outstanding checks totaling \$622 and an April 30 deposit in transit of \$1,898. During April, he recorded a \$280 check for the salary of a part-time employee as \$28. Stephens's Cash account shows an April 30 cash balance of \$1,936. How much cash does Stephens actually have at April 30?

L0 3

■ writing assignment

E5-24A. (Learning Objective 3: Making journal entries from a bank reconciliation) Use the data from Exercise 5-23A to make the journal entries that Stephens should record on April 30 to update his Cash account. Include an explanation for each entry.

L0 3

■ spreadsheet

E5-25A. (Learning Objective 3: Preparing a cash budget) Chad Communications, Inc., is preparing its cash budget for 20X7. Chad ended 20X6 with cash of \$82 million, and managers need to keep a cash balance of at least \$86 million for operations.

Collections from customers are expected to total \$11,308 million during 20X7, and payments for the cost of services and products should reach \$6,165 million. Operating expense payments are budgeted at \$2,543 million.

During 20X7, Chad expects to invest \$1,816 million in new equipment and sell older assets for \$116 million. Debt payments scheduled for 20X7 will total \$604 million. The company forecasts net income of \$895 million for 20X7 and plans to pay dividends of \$33 million.

Prepare Chad Communications' cash budget for 20X7. Will the budgeted level of cash receipts leave Chad with the desired ending cash balance of \$86 million, or will the company need additional financing? If so, how much?

L0 4

E5-26A. (Learning Objective 4: Reporting bad debts by the allowance method) On December 31, Darla's Travel has an accounts receivable balance of \$85,000. Allowance for Doubtful Accounts has a credit balance of \$680 before the year-end adjustment. Darla's Travel estimates that doubtful-account expense for the year is equal to 8% of ending receivables. Make the year-end entry to record doubtful-account expense. Show how the accounts receivable and the allowance for doubtful accounts are reported on the Balance Sheet.

L0 4

E5-27A. (Learning Objective 4: Using the allowance method for bad debts) On September 30, Hilly Mountain Party Planners had a \$35,000 balance in Accounts Receivable and a \$2,300 credit balance in Allowance for Uncollectible Accounts. During October, the store made credit sales of \$163,000. October collections on account were \$142,000, and write-offs of uncollectible receivables totaled \$3,200. The required allowance at year-end is calculated to be \$4,500.

Requirements

1. Journalize sales, collections, write-offs of uncollectibles, and uncollectible-account expense by the allowance method during October. Explanations are not required.
2. Show the ending balances in Accounts Receivable, Allowance for Uncollectible Accounts, and *Net Accounts Receivable* at October 31. How much does the store expect to collect?
3. Show how the store will report Accounts Receivable on its October 31 Balance Sheet.

L0 4

E5-28A. (Learning Objective 4: Using the aging approach to estimate bad debts) On December 31, before any year-end adjustments, the Accounts Receivable balance of Athens Company is \$260,000. The Allowance for Doubtful Accounts has a \$12,500 credit balance. Athens Company prepares the following aging schedule for Accounts Receivable:

Total Balance	Age of Accounts			
	1–30 Days	31–60 Days	61–90 Days	Over 90 Days
\$210,000	\$100,000	\$80,000	\$50,000	\$30,000
Estimated uncollectible	0.5%	2.0%	5.0%	50.0%

Requirements

■ spreadsheet

- Based on the aging of accounts receivable, is the unadjusted balance of the allowance account adequate? Too high? Too low?
- Make the entry required by the aging schedule. Prepare a T-account for the allowance.
- Show how Athens Company will report Accounts Receivable on its December 31 Balance Sheet.

E5-29A. (Learning Objective 4: Recording notes receivable and accruing interest revenue)

L0 4

Record the following note receivable transactions in the journal of Aegean Realty. How much interest revenue did Aegean earn this year? Use a 365-day year for interest computations, and round interest amounts to the nearest dollar.

Sept. 1	Loaned \$12,000 cash to Cristy Rocker on a one-year, 8% note.
Nov. 6	Performed service for Top Masters, receiving a 90-day, 6% note for \$10,000.
16	Received a \$5,000, six-month, 12% note on account from Veron, Inc.
30	Accrued interest revenue for the year.

E5-30A. (Learning Objective 5: Using the current ratio and days' sales in receivables to evaluate a company) Champi, Inc., reported the following items at December 31, 20X8 and 20X7:

L0 5

A1						
	A	B	C	D	E	F
1	Balance Sheets (Summarized)					
2		Year End			Year End	
3		20X8	20X7		20X8	20X7
4	Current assets:			Current liabilities:		
5	Cash	\$ 3,000	\$ 9,000	Accounts payable	\$ 18,000	\$ 22,000
6	Marketable securities	20,000	10,000	Other current liabilities	105,000	105,000
7	Accounts receivable, net	58,000	72,000	Long-term liabilities	15,000	19,000
8	Inventory	90,000	88,000			
9	Other current assets	2,000	2,000	Shareholders' equity	145,000	145,000
10	Long-term assets	110,000	110,000			
11	Total assets	\$283,000	\$291,000	Total liabilities and equity	\$283,000	\$291,000
12						
13	Income Statement (partial): 20X8					
14	Sales Revenue	\$750,000				
15						

Requirement

- Compute Champi's (a) current ratio and (b) days' sales in average receivables for 20X8. Evaluate each ratio value as strong or weak. Champi sells on terms of net 30 days.

E5-31A. (Learning Objective 5: Analyzing a company's financial statements) Modern Limited, the electronics and appliance chain, reported these figures in millions of dollars:

L0 5

	20X7	20X6
Net sales.....	\$578,000	\$607,000
Receivables at end of year	3,750	4,920

Requirements

1. Compute Modern's average collection period during 20X7.
2. Is Modern's collection period long or short? Viflex Networks takes 25 days to collect its average level of receivables. Domarko, the overnight shipper, takes 35 days. What causes Modern's collection period to be so different?

Group B

LO 1 2

■ writing assignment

E5-32B. (Learning Objectives 1, 2: Learning about fraud; identifying internal control weaknesses) Identify the internal control weakness in the following situations. State how the person can hurt the company.

- a. Josh Michaels works as a security guard at CITY parking in Cologne, Germany. Michaels has a master key to the cash box where customers pay for parking. Each night Michaels prepares the cash report that shows (a) the number of cars that parked on the lot and (b) the day's cash receipts. Lily Cooper, the CITY treasurer, checks Michaels's figures by multiplying the number of cars by the parking fee per car. Cooper then deposits the cash in the bank.
- b. Ara Ahern is the purchasing agent for Superior Golf Equipment. Ahern prepares purchase orders based on requests from division managers of the company. Ahern faxes the purchase order to suppliers who then ship the goods to Superior. Ahern receives each incoming shipment and checks it for agreement with the purchase order and the related invoice. She then routes the goods to the respective division managers and sends the receiving report and the invoice to the accounting department for payment.

LO 2

E5-33B. (Learning Objective 2: Identifying internal control strengths and weaknesses) The following situations describe two cash payment situations and two cash receipt situations. In each pair, one set of internal controls is better than the other. Evaluate the internal controls in each situation as strong or weak, and give the reason for your answer.

Cash payments:

- a. Sturdy Structures, Inc., policy calls for project supervisors to purchase the equipment needed for jobs. The supervisors then submit the paid receipts to the home office for reimbursement. This policy enables supervisors to get the equipment quickly and keep construction jobs moving.
- b. Milton Malcolm Construction policy calls for construction supervisors to request the equipment needed for their jobs. The home office then purchases the equipment and has it shipped to the construction site.

Cash receipts:

- a. At Carl Auto Parts, cash received by mail goes straight to the accountant, who debits Cash and credits Accounts Receivable to record the collections from customers. The Cramer accountant then deposits the cash in the bank.
- b. Cash received by mail at Clear Vision Eye Clinic goes to the mail room, where a mail clerk opens envelopes and totals the cash receipts for the day. The mail clerk forwards customer checks to the cashier for deposit in the bank and forwards the remittance slips to the accounting department for posting credits to customer accounts.

LO 1 2

■ writing assignment

E5-34B. (Learning Objectives 1, 2: Learning about fraud; correcting an internal control weakness) Simon Stuart served as executive director of Downtown Scanlon, an organization created to revitalize Scanlon. Over the course of 11 years Stuart embezzled €273,000. How did Stuart do it? He did it by depositing subscriber cash receipts in his own bank account, writing Downtown Scanlon checks to himself, and creating phony entities that Downtown Scanlon wrote checks to.

Downtown Scanlon was led by a board of directors comprised of civic leaders. Stuart's embezzlement went undetected until Downtown Scanlon couldn't pay its bills.

Give four ways Stuart's embezzlement could have been prevented.

E5-35B. (Learning Objective 2: Evaluating internal control over cash receipts) Ripley stores use point-of-sale terminals as cash registers. The register shows the amount of each sale, the cash received from the customer, and any change returned to the customer. The machine also produces a customer receipt but keeps no record of transactions. At the end of the day, the clerk counts the cash in the register and gives it to the cashier for deposit in the company bank account.

L0 2

Write a memo to convince the store manager that there is an internal control weakness over cash receipts. Identify the weakness that gives an employee the best opportunity to steal cash and state how to prevent such a theft.

E5-36B. (Learning Objective 2, 4: Evaluating internal control over cash payments) Long Range Golf Company manufactures a popular line of golf clubs. Long Range Golf employs 167 workers and keeps their employment records on time sheets that show how many hours the employee works each week. On Friday the shop foreman collects the time sheets, checks them for accuracy, and delivers them to the payroll department for preparation of paychecks. The treasurer signs the paychecks and returns the checks to the payroll department for distribution to the employees.

L0 2 4

Identify the main internal control weakness in this situation, state how the weakness can hurt Long Range Golf, and propose a way to correct the weakness.

E5-37B. (Learning Objective 3: Classifying bank reconciliation items) The following items appear on a bank reconciliation.

L0 3

Classify each item as (a) an addition to the book balance, (b) a subtraction from the book balance, (c) an addition to the bank balance, or (d) a subtraction from the bank balance.

1. ___ Outstanding checks
2. ___ Bank error: The bank credited our account for a deposit made by another bank customer.
3. ___ Service charge
4. ___ Deposits in transit
5. ___ NSF check
6. ___ Bank collection of a note receivable on our behalf
7. ___ Book error: We debited Cash for €300. The correct debit was €3,000.

E5-38B. (Learning Objective 3: Preparing a bank reconciliation) D.J. Hilton's checkbook and March bank statement show the following:

L0 3

Date	Check No.	Item	Check	Deposit	Balance
3/1					\$ 575
4	852	Art Cafe	\$ 30		545
9		Dividends received		\$ 135	680
13	853	General Tire Co.	35		645
14	854	QuickMobil	56		589
18	855	Cash	60		529
26	856	Woodway Baptist Church	80		449
28	857	Bent Tree Apartments	255		194
31		Paycheck		1,120	1,314

Balance		\$575
Add: Deposits		135
Debit checks:	No.	Amount
	852	\$30
	853	35
	854	65*
	855	60
		(190)
Other charges:		
NSF check		\$20
Service charge		10
		(30)
Balance		490

*This is the correct amount for check number 854.

Requirement

1. Prepare Hilton's bank reconciliation at March 31.

L0 3

E5-39B. (Learning Objective 3: Preparing a bank reconciliation) Harry Sparks operates a bowling alley. He has just received the monthly bank statement at September 30 from City National Bank, and the statement shows an ending balance of €565. Listed on the statement are an EFT rent collection of €345, a service charge of €9, two NSF checks totaling €135, and a €12 charge for printed checks. In reviewing his cash records, Sparks identifies outstanding checks totaling €639 and a September 30 deposit in transit of €1,970. During September, he recorded a €320 check for the salary of a part-time employee as €32. Sparks's Cash account shows a September 30 cash balance of €1,995. How much cash does Sparks actually have at September 30?

L0 3

■ writing assignment

E5-40B. (Learning Objective 3: Making journal entries from a bank reconciliation) Use the data from Exercise 5-39B to make the journal entries that Sparks should record on September 30 to update his Cash account. Include an explanation for each entry.

L0 3

■ spreadsheet

E5-41B. (Learning Objective 3: Preparing a cash budget) Fellow Communications, Inc., is preparing its cash budget for 20X7. Fellow ended 20X6 with cash of €85 million, and managers need to keep a cash balance of at least €82 million for operations.

Collections from customers are expected to total €11,317 million during 20X7, and payments for the cost of services and products should reach €6,184 million. Operating expense payments are budgeted at €2,546 million.

During 20X7, Fellow expects to invest €1,841 million in new equipment and sell older assets for €125 million. Debt payments scheduled for 20X7 will total €602 million. The company forecasts net income of €883 million for 20X7 and plans to pay dividends of €321 million.

Prepare Fellow Communications' cash budget for 20X7. Will the budgeted level of cash receipts leave Fellow with the desired ending cash balance of €82 million, or will the company need additional financing? If so, how much?

L0 4

E5-42B. (Learning Objective 4: Reporting bad debts by the allowance method) At December 31, Will's Travel has an accounts receivable balance of €90,000. Allowance for Doubtful Accounts has a credit balance of €700 before the year-end adjustment. Will's Travel estimates that doubtful-account expense for the year is equal to 8% of ending receivables. Make the December 31 entry to record doubtful-account expense. Show how the Accounts Receivable and the Allowance for Doubtful Accounts are reported on the Balance Sheet.

L0 4

E5-43B. (Learning Objective 4: Using the allowance method for bad debts) On April 30, Hilltop Party Planners had a €38,000 balance in Accounts Receivable and a €4,500 credit balance in Allowance for Uncollectible Accounts. During May, the store made credit sales of €146,000. May collections on account were €123,000, and write-offs of uncollectible receivables totaled €2,300. Hilltop Party Planners estimates the required allowance for uncollectible accounts for the year is €4,000.

Requirements

1. Journalize sales, collections, write-offs of uncollectibles, and uncollectible-account expense by the allowance method during May. Explanations are not required.
2. Show the ending balances in Accounts Receivable, Allowance for Uncollectible Accounts, and *Net* Accounts Receivable at May 31. How much does the store expect to collect?
3. Show how the store will report Accounts Receivable on its May 31 Balance Sheet.

L0 4

■ spreadsheet

E5-44B. (Learning Objective 4: Using the aging approach to estimate bad debts) At December 31, before any year-end adjustments, the accounts receivable balance of Delta Electronics Company is €180,000. The allowance for doubtful accounts has a €6,500 credit balance. Delta Electronics Company prepares the following aging schedule for accounts receivable:

Total Balance	Age of Accounts			
	1–30 Days	31–60 Days	61–90 Days	Over 90 Days
\$180,000	€80,000	€50,000	€30,000	€10,000
Estimated uncollectible	0.5%	3.0%	7.0%	40.0%

Requirements

- Based on the aging of accounts receivable, is the unadjusted balance of the allowance account adequate? Too high? Too low?
- Make the entry required by the aging schedule. Prepare a T-account for the allowance.
- Show how Delta Electronics Company will report Accounts Receivable on its December 31 Balance Sheet.

E5-45B. (Learning Objective 4: Recording notes receivable and accruing interest revenue)

LO 4

Record the following note receivable transactions in the journal of Celtic Realty. How much interest revenue did Celtic earn this year? Use a 365-day year for interest computations, and round interest amounts to the nearest dollar.

Apr. 1	Loaned \$18,000 cash to Boyle Denvers on a one-year 10% note.
Jun. 6	Performed service for Pot Masters, receiving a 90-day, 8% note for \$15,000.
16	Received a \$4,000, six-month, 12% note on account from Veron, Inc.
30	Accrued interest revenue for the year.

E5-46B. (Learning Objective 5: Using the current ratio and days' sales in receivables to evaluate a company)

LO 5

Navajo, Inc., reported the following items at December 31, 20X8 and 20X7:

A1						
	A	B	C	D	E	F
1	Balance Sheets (Summarized)					
2		Year End			Year End	
3		20X8	20X7		20X8	20X7
4	Current assets:			Current liabilities:		
5	Cash	\$ 4,000	\$ 10,000	Accounts payable	\$ 17,000	\$ 18,000
6	Marketable securities	24,000	13,000	Other current liabilities	106,000	109,000
7	Accounts receivable, net	58,000	75,000	Long-term liabilities	15,000	19,000
8	Inventory	92,000	88,000			
9	Other current assets	6,000	6,000	Stockholders' equity	146,000	146,500
10	Long-term assets	100,000	100,000			
11	Total assets	\$284,000	\$292,000	Total liabilities and equity	\$284,000	\$292,000
12						
13	Income Statement (partial):	20X8				
14	Sales Revenue	\$730,000				
15						

Requirement

- Compute Navajo's (a) current ratio and (b) days' sales in average receivables for 20X8. Evaluate each ratio value as strong or weak. Navajo sells on terms of net 30 days.

E5-47B. (Learning Objective 5: Analyzing a company's financial statements)

LO 5

Limited, the electronics and appliance chain, reported these figures in millions of dollars:

	20X7	20X6
Net sales.....	€574,000	€603,000
Receivables at end of year	3,670	4,380

Requirements

1. Compute Contemporary's average collection period during 20X7.
2. Is Contemporary's collection period long or short? Kurzweil Networks takes 40 days to collect its average level of receivables. Damascus, the overnight shipper, takes 25 days. What causes Contemporary's collection period to be so different?

Challenge Exercises

L0 1 2

E5-48. (Learning Objectives 1, 2: Learning about fraud; evaluating internal controls over cash payments; focusing on ethical considerations) Shirley Hopper, the owner of Shirley's Perfect Presents, has delegated management of the business to Leslie Temple, a friend. Hopper drops by to meet customers and check up on cash receipts, but Temple buys the merchandise and handles cash payments. Business has been very good lately, and cash receipts have kept pace with the apparent level of sales. However, for a year or so, the amount of cash on hand has been too low. When asked about this, Temple explains that suppliers are charging more for goods than in the past. During the past year, Temple has taken two expensive vacations, and Hopper wonders how Temple can afford these trips on her \$60,000 annual salary and commissions.

List at least three ways Temple could be defrauding Hopper of cash. In each instance also identify how Hopper can determine whether Temple's actions are ethical. Limit your answers to the store's cash payments. The business pays all suppliers by check (no EFTs).

L0 3

E5-49. (Learning Objective 3: Preparing and using a cash budget) Dexter Dunne, the chief financial officer, is responsible for The Furniture Mart's cash budget for 20X6. The budget will help Dunne determine the amount of long-term borrowing needed to end the year with a cash balance of \$160,000. Dunne's assistants have assembled budget data for 20X6, which the computer printed in alphabetical order. Not all the data items reproduced below are used in preparing the cash budget.

(Assumed Data)	(In thousands)
Actual cash balance, December 31, 20X5	\$ 150
Budgeted total assets	24,377
Budgeted total current assets	7,769
Budgeted total current liabilities	4,620
Budgeted total liabilities	12,088
Budgeted total shareholders' equity	12,289
Collections from customers	22,100
Dividend payments	371
Issuance of shares	647
Net income	1,183
Payment of long-term and short-term debt	984
Payment of operating expenses	2,438
Purchases of inventory items	14,655
Purchase of property and equipment	1,257

Requirements

1. Prepare the cash budget of The Furniture Mart, Inc.
2. Compute The Furniture Mart's budgeted current ratio at December 31, 20X6. Based on the current ratio and on the cash budget, would you lend \$100,000 to The Furniture Mart? Give the reason for your decision.

E5-50. (Learning Objective 4: Reconstructing receivables and bad-debt amounts) Suppose Hearts, Inc., reported net receivables of \$2,856 million and \$2,628 million at January 31, 20X7 and 20X6, after subtracting allowances of \$74 million and \$69 million on these respective dates. Hearts earned total revenue of \$54,333 million (all on account) and recorded doubtful-account expense of \$18 million for the year ended January 31, 20X7.

LO 4

Requirement

1. Use this information to measure the following amounts for the year ended January 31, 20X7:
 - a. Write-offs of uncollectible receivables.
 - b. Collections from customers.

Quiz

Test your understanding of internal control and cash by answering the following questions. Answer each question by selecting the best choice from among the answers given.

Q5-51. All of the following are objectives of internal control *except*

- a. to safeguard assets.
- b. to maximize net income.
- c. to comply with legal requirements.
- d. to ensure accurate and reliable accounting records.

Q5-52. Requiring that an employee with no access to cash do the accounting is an example of which characteristic of internal control?

- | | |
|-------------------------------------|---------------------------|
| a. Competent and reliable personnel | c. Separation of duties |
| b. Assignment of responsibility | d. Monitoring of controls |

Q5-53. All of the following are controls for cash received over the counter *except*

- a. the cash drawer should open only when the sales clerk enters an amount on the keys.
- b. the customer should be able to see the amounts entered into the cash register.
- c. the sales clerk must have access to the cash register tape.
- d. a printed receipt must be given to the customer.

Q5-54. If a bookkeeper mistakenly recorded a \$45 deposit as \$54, the error would be shown on the bank reconciliation as a

- | | |
|---|--|
| a. \$9 deduction from the book balance. | c. \$54 deduction from the book balance. |
| b. \$9 addition to the book balance. | d. \$54 addition to the book balance. |

Q5-55. If a bank reconciliation included a deposit in transit of \$780, the entry to record this reconciling item would include

- | | |
|--------------------------------|---|
| a. a debit to cash for \$780. | c. a credit to prepaid insurance for \$780. |
| b. a credit to cash for \$780. | d. Nothing, as no entry is required. |

Q5-56. Before paying an invoice for goods received on account, the controller or treasurer should ensure that

- a. the company is paying for the goods it ordered.
- b. the company has not already paid this invoice.
- c. the company is paying for the goods it actually received.
- d. all of the above are taken into account.

Q5-57. Under the allowance method for uncollectible receivables, the entry to record uncollectible-account expense has what effect on the financial statements?

- a. Decreases assets and has no effect on net income
- b. Decreases net income and decreases assets
- c. Decreases owners' equity and increases liabilities
- d. Increases expenses and increases owners' equity

Q5-58. Van Gogh Company uses the aging method to adjust the allowance for uncollectible accounts at the end of the period. At December 31, the balance of accounts receivable is \$240,000 and the allowance for uncollectible accounts has a credit balance of \$5,000 (before adjustment). An analysis of accounts receivable produced the following age groups:

Current	\$180,000
60 days past due.....	50,000
Over 60 days past due.....	10,000
	<u>\$240,000</u>

Based on past experience, Van Gogh estimates that the percentage of accounts that will prove to be uncollectible within the three age groups is 2%, 8%, and 20%, respectively. Based on these facts, the adjusting entry for uncollectible accounts should be made in the amount of

- a. \$9,600.
- b. \$5,000.
- c. \$4,600.
- d. \$14,600.

Q5-59. Refer to Question 5-58. The net receivables on the Balance Sheet is

- a. \$230,400.
- b. \$235,000.
- c. \$235,400.
- d. \$225,400.

Q5-60. Gordon Company uses the aging method in setting its allowance for doubtful receivables. Allowance for doubtful accounts prior to adjustment has a credit balance of \$2,900. Management estimates that due to the economic crisis, a higher level of allowance is necessary and decides that a \$5,600 allowance is an appropriate amount at the year-end. The amount of expense to report on the Income Statement will be

- a. \$2,700.
- b. \$5,600.
- c. \$8,500.
- d. \$2,900.

Problems MyLab Accounting

Select Group A and Group B problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting.

Group A

P5-61A. (*Learning Objectives 1, 2: Learning about fraud; identifying internal control weakness*) Each of the following situations reveals an internal control weakness:

- a. In evaluating the internal control over cash payments of Bermingham Manufacturing, an auditor learns that the purchasing agent is responsible for purchasing diamonds for use in the company's manufacturing process, approving the invoices for payment, and signing the checks. No supervisor reviews the purchasing agent's work.
- b. Mary Angela owns an architectural firm. Angela's staff consists of 20 professional architects, and Angela manages the office. Often, Angela's work requires her to travel to meet with clients. During the past six months, Angela has observed that when she returns from a business trip, the architecture jobs in the office have not progressed satisfactorily. Angela learns that when she is away, three of her senior architects take over office management and neglect their normal duties. One employee could manage the office.
- c. J. T. Derby has been an employee of the City of Maron for many years. Because the city is small, Derby performs all accounting duties, plus opening the mail, preparing the bank deposit, and preparing the bank reconciliation.

Requirements

1. Identify the missing internal control characteristic in each situation.
2. Identify each firm's possible problem.
3. Propose a solution to the problem.

P5-62A. (*Learning Objective 3: Using the bank reconciliation as a control device*) The cash data of Dursley Automotive for July 20X6 follow:

L0 1 2

■ writing assignment

L0 3

■ writing assignment

Cash					Account No. 101
Date	Item	Jrnl. Ref.	Debit	Credit	Balance
Jul. 1	Balance				7,900
31		CR6	9,154		17,054
31		CP11		9,097	7,957

Cash Receipts (CR)		Cash Payments (CP)	
Date	Cash Debit	Check No.	Cash Credit
Jul. 2	\$2,771	3113	\$1,503
8	516	3114	1,149
10	1,682	3115	1,620
16	871	3116	19
22	352	3117	825
29	934	3118	91
30	2,028	3119	450
Total	<u>\$9,154</u>	3120	975
		3121	215
		3122	<u>2,250</u>
		Total	<u>\$9,097</u>

Dursley received the following bank statement on July 31, 20X6:

■ spreadsheet

A1			
	A	B	C
1	Bank Statement for July 20X6		
2	Beginning balance		\$ 7,900
3	Deposits and other additions:		
4	July 1	\$ 760 EFT	
5	4	2,771	
6	9	516	
7	12	1,682	
8	17	871	
9	22	352	
10	23	1,270 BC	8,222
11	Checks and other deductions:		
12	July 7	\$1,503	
13	13	1,260	
14	14	417 US	
15	15	1,149	
16	18	19	
17	21	343 EFT	
18	26	825	
19	30	91	
20	30	28 SC	(5,635)
21	Ending balance		<u>\$10,487</u>
22			

Explanation: BC—bank collection, EFT—electronic funds transfer, US—unauthorized signature, SC—service charge

Additional data for the bank reconciliation include the following:

- The EFT deposit was a receipt of monthly rent. The EFT debit was a monthly insurance payment.
- The unauthorized signature check was received from a customer.
- The correct amount of check number 3115, a payment on account, is \$1,260. (Dursley's accountant mistakenly recorded the check for \$1,620.)

Requirements

1. Prepare the Dursley Automotive bank reconciliation at July 31, 20X6.
2. Describe how a bank account and the bank reconciliation help the general manager control Dursley's cash.

L0 2

■ spreadsheet

P5-63A. (Learning Objective 2: Identifying internal control weakness in sales and cash receipts) Smooth Skin Care makes all sales on credit. Cash receipts arrive by mail, usually within 30 days of the sale. Kate Marco opens envelopes and separates the checks from the accompanying remittance advices. Marco forwards the checks to another employee, who makes the daily bank deposit but has no access to the accounting records. Marco sends the remittance advices, which show the amount of cash received, to the accounting department for entry in the accounts receivable. Marco's only other duty is to grant allowances to customers. (An allowance decreases the amount that the customer must pay.) When Marco receives a customer check for less than the full amount of the invoice, she records the allowance in the accounting records and forwards the document to the accounting department.

Requirement

1. You are a new employee of Smooth Skin Care. Write a memo to the company president identifying the internal control weakness in this situation. State how to correct the weakness.

L0 3

■ writing assignment

P5-64A. (Learning Objective 3: Preparing a bank reconciliation and the related journal entries) The August 31 bank statement of Grayson Engineering Associates has just arrived from Carolina First Bank. To prepare the Grayson bank reconciliation, you gather the following data:

- a. Grayson's Cash account shows a balance of \$8,158.71 on August 31.
- b. The August 31 bank balance is \$8,789.23.
- c. The bank statement shows that Grayson earned \$18.58 of interest on its bank balance during August. This amount was added to Grayson's bank balance.
- d. Grayson pays utilities (\$740) and insurance (\$290) by EFT.
- e. The following Grayson checks did not clear the bank by August 31:

Check No.	Amount
237	\$401.00
288	74.82
291	33.25
293	165.55
294	236.00
295	47.75
296	107.85

- f. The bank statement includes a deposit of \$895.15, collected on account by the bank on behalf of Grayson.
- g. The bank statement lists a \$7.50 bank service charge.
- h. On August 31, the Grayson treasurer deposited \$354.38, which will appear on the September bank statement.
- i. The bank statement includes a \$268.40 deposit that Grayson did not make. The bank added \$268.40 to Grayson's account for another company's deposit.
- j. The bank statement includes two charges for returned checks from customers. One is a \$158.50 check received from a customer with the imprint "Unauthorized Signature." The other is a nonsufficient funds check in the amount of \$67.45 received from another customer.

Requirements

1. Prepare the bank reconciliation for Grayson Engineering Associates.
2. Journalize the August 31 transactions needed to update Grayson's Cash account. Include an explanation for each entry.

P5-65A. (Learning Objective 3: Preparing a cash budget and using cash-flow information) John Watson, chief financial officer of Scott Wireless, is responsible for the company's budgeting process. Watson's staff is preparing the Scott cash budget for 20X7. A key input to the budgeting process is last year's statement of cash flows, which follows (amounts in thousands):

L0 3

A1		
	A	B
1	Scott Wireless Statement of Cash Flows 20X6	
2	(In thousands)	
3	Cash Flows from Operating Activities	
4	Collections from customers	\$ 64,000
5	Interest received	300
6	Purchases of inventory	(49,000)
7	Operating expenses	(13,500)
8	Net cash provided by operating activities	1,800
9	Cash Flows from Investing Activities	
10	Purchases of equipment	(4,800)
11	Purchases of investments	(400)
12	Sales of investments	500
13	Net cash used for investing activities	(4,700)
14	Cash Flows from Financing Activities	
15	Payment of long-term debt	(400)
16	Issuance of shares	1,700
17	Payment of cash dividends	(300)
18	Net cash provided by financing activities	1,000
19	Cash	
20	Increase (decrease) in Cash	(1,900)
21	Cash, beginning of year	3,400
22	Cash, end of year	\$ 1,500
23		

Requirements

1. Prepare the Scott Wireless cash budget for 20X7. Date the budget simply “20X7” and denote the beginning and ending cash balances as “beginning” and “ending.” Assume the company expects 20X7 to be the same as 20X6, but with the following changes:
 - a. In 20X7, the company expects a 12% increase in collections from customers and a 25% increase in purchases of inventory.
 - b. There will be no sales of investments in 20X7.
 - c. Scott plans to issue no shares in 20X7.
 - d. Scott plans to end the year with a cash balance of \$3,500 (thousand).
2. Does the company’s cash budget for 20X7 suggest that Scott is growing, holding steady, or decreasing in size? (Challenge)

P5-66A. (*Learning Objective 4: Accounting for revenue, collections, and uncollectibles*)

This problem takes you through the accounting for sales, receivables, and uncollectibles for Mail Time Corp., the overnight shipper. By selling on credit, the company cannot expect to collect 100% of its accounts receivable. At May 31, 20X6 and 20X7, respectively, Mail Time Corp. reported the following on its Balance Sheet (in millions of dollars):

LO 4

■ writing assignment

	May 31	
	20X7	20X6
Accounts receivable.....	\$3,703	\$3,345
Less: Allowance for uncollectible accounts.....	<u>(139)</u>	<u>(166)</u>
Accounts receivable, net.....	\$3,564	\$3,179

During the year ended May 31, 20X7, Mail Time Corp. earned service revenue and collected cash from customers. Assume that Mail Time wrote off uncollectible receivables. At year-end Mail Time ended with the foregoing May 31, 20X7, balances.

Requirements

1. Prepare T-accounts for Accounts Receivable and Allowance for Uncollectibles and insert the May 31, 20X6, balances as given.
2. Journalize the following assumed transactions of Mail Time Corp. for the year ended May 31, 20X7 (explanations are not required):
 - a. Service revenue on account, \$32,491 million
 - b. Collections from customers on account, \$31,779 million
 - c. Write-offs of uncollectible accounts receivable, \$354 million
 - d. Uncollectible-account expense, \$327 million
3. Post your entries to the Accounts Receivable and the Allowance for Uncollectibles T-accounts.
4. Compute the ending balances for the two T-accounts and compare your balances to the actual May 31, 20X7, amounts. They should be the same.
5. Show what Mail Time would report on its Income Statement for the year ended May 31, 20X7.

L0 4

P5-67A. (Learning Objective 4: Using the aging approach for uncollectibles) The September 30, 20X7, records of Effective Communications include these accounts:

Accounts Receivable.....	\$270,000
Allowance for Doubtful Accounts.....	(9,100)

At year-end (December 31), the company ages its receivables and adjusts the balance in Allowance for Doubtful Accounts to correspond to the aging schedule. During the last quarter of 20X7, the company completed the following selected transactions:

Nov. 30	Wrote off as uncollectible the \$1,100 account receivable from Black Carpets and the \$500 account receivable from Old Timer Antiques.
Dec. 31	Adjusted the Allowance for Doubtful Accounts and recorded doubtful-account expense at year-end, based on the aging of receivables, which follows.

Accounts Receivable	Age of Accounts			
	1–30 Days	31–60 Days	61–90 Days	Over 90 Days
\$235,000	\$150,000	\$35,000	\$20,000	\$30,000
Estimated percent uncollectible	0.2%	2.0%	8.0%	40.0%

Requirements

1. Record the transactions in the journal. Explanations are not required.
2. Prepare a T-account for Allowance for Doubtful Accounts and post to that account.
3. Show how Effective Communications will report its accounts receivable on its Balance Sheet at December 31, 20X7.

L0 4

P5-68A. (Learning Objective 4: Accounting for notes receivable and accrued interest revenue) Healthy Meal completed the following selected transactions.

20X6

- Oct. 31 Sold goods to Buy Low Foods, receiving a \$36,000, three-month, 6.50% note.
 Dec. 31 Made an adjusting entry to accrue interest on the Buy Low Foods note.

20X7

- Jan. 31 Collected the Buy Low Foods note.
 Feb. 18 Received a 90-day, 8.25%, \$7,400 note from Dutton Market on account.
 19 Sold the Dutton Market note to Amherst Bank, receiving cash of
 \$7,300. (Debit the difference to financing expense.)
 Nov. 11 Lent \$16,400 cash to Street Provisions, receiving a 90-day, 9.50% note.
 Dec. 31 Accrued the interest on the Street Provisions note.

Requirements

- Record the transactions in Healthy Meal's journal. Round interest amounts to the nearest dollar. Explanations are not required.
- Show what Healthy Meal will report on its comparative classified Balance Sheet at December 31, 20X7, and December 31, 20X6.

P5-69A. (*Learning Objective 5: Using ratio data to evaluate a company's financial position*)

The comparative financial statements of Highland Pools, Inc., for 20X7, 20X6, and 20X5 included the following select data:

L0 5■ **spreadsheet**

	(In millions)		
	20X7	20X6	20X5
Balance Sheet			
Current assets:			
Cash.....	\$ 80	\$ 70	\$ 60
Short-term investments	145	170	120
Receivables, net of allowance for doubtful accounts of \$7, \$6, and \$4, respectively	290	260	250
Inventories	355	345	300
Prepaid expenses	60	30	55
Total current assets	<u>\$ 930</u>	<u>\$ 875</u>	<u>\$ 785</u>
Total current liabilities.....	\$ 590	\$ 640	\$ 690
Income Statement			
Net sales	\$5,900	\$5,150	\$4,230

Requirements

- Compute current ratio and days' sales in receivables for 20X7 and 20X6:
- Which ratio(s) improved from 20X6 to 20X7 and which ratio(s) deteriorated? Is this trend favorable or unfavorable?

Group B

P5-70B. (*Learning Objectives 1, 2: Learning about fraud; identifying internal control weakness*) Each of the following situations reveals an internal control weakness:

L0 1 2■ **writing assignment**

Situation a. In evaluating the internal control over cash payments of Chester Manufacturing, an auditor learns that the purchasing agent is responsible for purchasing diamonds for use in the company's manufacturing process, approving the invoices for payment, and signing the checks. No supervisor reviews the purchasing agent's work.

Situation b. Rose White owns an architectural firm. White's staff consists of 15 professional architects, and White manages the office. Often, White's work requires her to travel to meet with clients. During the past six months, White has observed that when she returns from a business trip, the architecture jobs in the office have not progressed satisfactorily. White learns that when she is away, two of her senior architects take over office management and neglect their normal duties. One employee could manage the office.

Situation c. M. J. Davis has been an employee of the City of Northport for many years. Because the city is small, Davis performs all accounting duties, plus opening the mail, preparing the bank deposit, and preparing the bank reconciliation.

Requirements

1. Identify the missing internal control characteristic in each situation.
2. Identify each firm's possible problem.
3. Propose a solution to the problem.

L0 3

■ writing assignment

■ spreadsheet

P5-71B. (*Learning Objective 3: Using the bank reconciliation as a control device*) The cash data of Dudley Automotive for January 20X6 follow:

Cash					Account No. 101
Date	Item	Jrnl. Ref.	Debit	Credit	Balance
Jan. 1	Balance				7,200
31		CR 6	9,156		16,356
31		CP 11		10,023	6,333

Cash Receipts (CR)		Cash Payments (CP)	
Date	Cash Debit	Check No.	Cash Credit
Jan. 2	\$2,726	3113	\$ 1,475
8	572	3114	1,925
10	1,647	3115	1,530
16	837	3116	32
22	436	3117	870
29	865	3118	132
30	2,073	3119	483
Total	<u>\$9,156</u>	3120	995
		3121	249
		3122	2,332
		Total	<u>\$10,023</u>

Dudley received the following bank statement on January 31, 20X6:

A1			
	A	B	C
1	Bank Statement for January 20X6		
2	Beginning balance		€ 7,200
3	Deposits and other additions:		
4	Jan. 1	\$ 680 EFT	
5	4	2,726	
6	9	572	
7	12	1,647	
8	17	837	
9	22	436	
10	23	1,340 BC	8,238
11	Checks and other deductions:		
12	Jan. 7	€1,475	
13	13	1,350	
14	14	436 US	
15	15	1,925	
16	18	32	
17	21	313 EFT	
18	26	870	
19	30	132	
20	30	22 SC	(6,555)
21	Ending balance		€ 8,883
22			

Explanation: BC—bank collection, EFT—electronic funds transfer, US—unauthorized signature, SC—service charge

Additional data for the bank reconciliation include the following:

- The EFT deposit was a receipt of monthly rent. The EFT debit was a monthly insurance expense.
- The unauthorized signature check was received from a customer.
- The correct amount of check number 3115, a payment on account, is €1,350. (Dudley's accountant mistakenly recorded the check for €1,530.)

Requirements

- Prepare the Dudley Automotive bank reconciliation at January 31, 20X6.
- Describe how a bank account and the bank reconciliation help the general manager control Dudley's cash.

P5-72B. (*Learning Objective 2: Identifying internal control weakness in sales and cash receipts*) Glowing Skin Care makes all sales on credit. Cash receipts arrive by mail, usually within 30 days of the sale. Emily Nolan opens envelopes and separates the checks from the accompanying remittance advices. Nolan forwards the checks to another employee, who makes the daily bank deposit but has no access to the accounting records. Nolan sends the remittance advices, which show the amount of cash received, to the accounting department for entry in the accounts receivable. Nolan's only other duty is to grant allowances to customers. (An *allowance* decreases the amount that the customer must pay.) When Nolan receives a customer check for less than the full amount of the invoice, she records the allowance in the accounting records and forwards the document to the accounting department.

L0 2

writing assignment

Requirement

1. You are a new employee of Glowing Skin Care. Write a memo to the company president identifying the internal control weakness in this situation. State how to correct the weakness.

L0 3

■ spreadsheet

P5-73B. (Learning Objective 3: Preparing a bank reconciliation and the related journal entries) The October 31 bank statement of Drake Engineering Associates has just arrived from Carolina First Bank. To prepare the Drake bank reconciliation, you gather the following data:

- a. Drake's Cash account shows a balance of €7,506.86 on October 31.
- b. The October 31 bank balance is €8,353.87.
- c. The bank statement shows that Drake earned €15.65 of interest on its bank balance during October. This amount was added to Drake's bank balance.
- d. Drake pays utilities (€780) and insurance (€260) by EFT.
- e. The following Drake checks did not clear the bank by October 31:

Check No.	Amount
237	€403.15
288	78.98
291	36.39
293	155.45
294	234.00
295	47.50
296	106.79

- f. The bank statement includes a deposit of €925.20, collected on account by the bank on behalf of Drake.
- g. The bank statement lists a €8.25 bank service charge.
- h. On October 31, the Drake treasurer deposited €350.80, which will appear on the November bank statement.
- i. The bank statement includes a €496.10 deposit that Drake did not make. The bank added €496.10 to Drake's account for another company's deposit.
- j. The bank statement includes two charges for returned checks from customers. One is a €185.50 check received from a customer with the imprint "Unauthorized Signature." The other is a nonsufficient funds check in the amount of €67.65 received from another customer.

Requirements

1. Prepare the bank reconciliation for Drake Engineering Associates.
2. Journalize the October 31 transactions needed to update Drake's Cash account. Include an explanation for each entry.

L0 3

P5-74B. (Learning Objective 3: Preparing a cash budget and using cash-flow information) Don Beecher, chief financial officer of Carlisle Wireless, is responsible for the company's budgeting process. Beecher's staff is preparing the Carlisle cash budget for 20X7. A key input to the budgeting process is last year's statement of cash flows, which follows (amount in thousands):

A1		
	A	B
1	Carlisle Wireless Statement of Cash Flows 20X6	
2	(In thousands)	
3	Cash Flows from Operating Activities	
4	Collections from customers	€ 62,000
5	Interest received	700
6	Purchases of inventory	(47,000)
7	Operating expenses	(13,700)
8	Net cash provided by operating activities	2,000
9	Cash Flows from Investing Activities	
10	Purchases of equipment	(4,100)
11	Purchases of investments	(300)
12	Sales of investments	900
13	Net cash used for investing activities	(3,500)
14	Cash Flows from Financing Activities	
15	Payment of long-term debt	(500)
16	Issuance of shares	1,500
17	Payment of cash dividends	(400)
18	Net cash provided by financing activities	600
19	Cash	
20	Increase (decrease) in Cash	(900)
21	Cash, beginning of year	2,800
22	Cash, end of year	€ 1,900
23		

Requirements

- Prepare the Carlisle Wireless cash budget for 20X7. Date the budget simply “20X7” and denote the beginning and ending cash balances as “beginning” and “ending.” Assume the company expects 20X7 to be the same as 2010, but with the following changes:
 - In 20X7, the company expects a 15% increase in collections from customers and a 22% increase in purchases of inventory.
 - There will be no sales of investments in 20X7.
 - Carlisle plans to issue no shares in 20X7.
 - Carlisle plans to end the year with a cash balance of €3,650 (thousand).
- Does the company’s cash budget for 20X7 suggest that Carlisle is growing, holding steady, or decreasing in size?

P5-75B. (Learning Objective 4: Accounting for revenue, collections, and uncollectibles)

This problem takes you through the accounting for sales, receivables, and uncollectibles for On-time Delivery Corp, the overnight shipper. By selling on credit, the company cannot expect to collect 100% of its accounts receivable. At May 31, 20X6 and 20X7, respectively, On-time Delivery Corp. reported the following on its Balance Sheet (in millions of euros):

L0 4

	May 31	
	20X7	20X6
Accounts receivable.....	€3,693	€3,435
Less: Allowance for uncollectible accounts.....	(129)	(156)
Accounts receivable, net	€3,564	€3,279

During the year ended May 31, 20X7, On-time Delivery Corp. earned sales revenue and collected cash from customers. Assume that On-time Delivery Corp. wrote off uncollectible receivables. At year-end, On-time Delivery Corp. ended with the foregoing May 31, 20X7, balances.

Requirements

1. Prepare T-accounts for Accounts Receivable and Allowance for Uncollectibles, and insert the May 31, 20X6, balances as given.
2. Journalize the following transactions of On-time Delivery for the year ended May 31, 20X7. (Explanations are not required.)
 - a. Service revenue on account, €32,587 million.
 - b. Collections from customers on account, €31,979 million.
 - c. Write-offs of uncollectible accounts receivable, €350 million.
 - d. Uncollectible-account expense, €323 million.
3. Post to the Accounts Receivable and Allowance for Uncollectibles T-accounts.
4. Compute the ending balances for the two T-accounts and compare your balances to the actual May 31, 20X7, amounts. They should be the same.
5. Show what On-time Delivery should report on its Income Statement for the year ended May 31, 20X7.

L0 4

P5-76B. (*Learning Objective 4: Using the aging approach for uncollectibles*) The September 30, 20X7, records of Image Communications include these accounts:

Accounts receivable.....	€290,000
Allowance for doubtful accounts.....	(9,700)

At year-end, the company ages its receivables and adjusts the balance in Allowance for Doubtful Accounts to correspond to the aging schedule. During the last quarter of 20X7, the company completed the following selected transactions:

Dec. 28	Wrote off as uncollectible the €1,300 account receivable from Blue Carpets and the €700 account receivable from Show-N-Tell Antiques.
Dec. 31	Adjusted the Allowance for Doubtful Accounts and recorded doubtful-account expense at year-end, based on the aging of receivables, which follows.

Total Balance	Age of Accounts			
	1–30 Days	31–60 Days	61–90 Days	Over 90 Days
€238,000	€160,000	€40,000	€18,000	€20,000
Estimated uncollectible	0.5%	1.0%	10.0%	50.0%

Requirements

1. Record the transactions in the journal. Explanations are not required.
2. Prepare a T-account for Allowance for Doubtful Accounts and post to that account.
3. Show how Image Communications will report its accounts receivable on its Balance Sheet at December 31, 20X7.

L0 4

P5-77B. (*Learning Objective 4: Accounting for notes receivable and accrued interest revenue*) Organic Meals completed the following selected transactions:

20X6

- Nov. 30 Sold goods to Bragg Market, receiving a €30,000, three-month, 4.25% note.
 Dec. 31 Made an adjusting entry to accrue interest on the Bragg Market note.

20X7

- Feb. 28 Collected the Bragg Market note.
 Mar. 1 Received a 90-day, 8.50%, €7,500 note from Don's Market on account.
 1 Sold the Don's Market note to Chelmsford Bank, receiving cash of
 €7,400. (Debit the difference to financing expense.)
 Dec. 16 Lent €14,400 cash to Stratford Provisions, receiving a 90-day, 10.00% note.
 Dec. 31 Accrued the interest on the Stratford Provisions note.

Requirements

- Record the transactions in Organic Meals' journal. Round all amounts to the nearest dollar. Explanations are not required.
- Show what Organic Meals will report on its comparative classified Balance Sheet at December 31, 20X7, and December 31, 20X6.

P5-78B. (*Learning Objective 5: Using ratio data to evaluate a company's financial position*) The comparative financial statements of Gold Pools, Inc., for 20X7, 20X6, and 20X5 included the following select data:

L0 5

■ spreadsheet

	(In millions)		
	20X7	20X6	20X5
Balance Sheet			
Current assets:			
Cash.....	€ 70	€ 80	€ 50
Short-term investments	145	160	110
Receivables, net of allowance for doubtful accounts of €7, €6, and €4, respectively	260	230	200
Inventories	360	345	310
Prepaid expenses	70	10	40
Total current assets	€ 905	€ 825	€ 710
Total current liabilities.....	€ 550	€ 600	€ 660
Income Statement			
Net sales	€5,890	€5,120	€4,210

Requirements

- Compute current ratio and days' sales in receivables for 20X7 and 20X6.
- Which ratio(s) improved from 20X6 to 20X7 and which ratio(s) deteriorated? Is this trend favorable or unfavorable?

APPLY YOUR KNOWLEDGE**Decision Cases**

Case 1. (*Learning Objectives 1, 2, 3: Learning about fraud; using a bank reconciliation to detect a theft*) Environmental Concerns, Inc., has poor internal control. Recently, Oscar Benz, the manager, has suspected the bookkeeper of stealing. Details of the business's cash position at September 30 follow.

L0 1 2 3

- a. The Cash account shows a balance of \$10,912. This amount includes a September 30 deposit of \$3,994 that does not appear on the September 30 bank statement.
- b. The September 30 bank statement shows a balance of \$8,324. The bank statement lists a \$200 bank collection, a \$13 service charge, and a \$41 NSF check. The accountant has not recorded any of these items.
- c. At September 30, the following checks are outstanding:

Check No.	Amount
154	\$116
256	150
278	853
291	990
292	206
293	145

- d. The bookkeeper receives all incoming cash and makes the bank deposits. He also reconciles the monthly bank statement. Here is his September 30 reconciliation:

Balance per books, September 30	\$10,912
Add: Outstanding check	1,260
Bank collection	200
Subtotal	12,372
Less: Deposits in transit	\$3,994
Service charge	13
NSF check	41
Balance per bank, September 30	<u>\$ 8,324</u>

Requirement

1. Benz has requested that you determine whether the bookkeeper has stolen cash from the business and, if so, how much. He also asks you to explain how the bookkeeper attempted to conceal the theft. To make this determination, you perform a proper bank reconciliation. There are no bank or book errors. Benz also asks you to evaluate the internal controls and to recommend any changes needed to improve them.

L0 1 2

Case 2. (*Learning Objectives 1, 2: Learning about fraud; correcting an internal control weakness*) This case is based on an actual situation experienced by one of the authors. Gilead Construction, headquartered in Topeka, Kansas, built a motel in Kansas City. The construction foreman, Slim Pickins, hired the workers for the project. Pickins had his workers fill out the necessary tax forms and sent the employment documents to the home office.

Work on the motel began on May 1 and ended in December. Each Thursday evening, Pickins filled out a time card that listed the hours worked by each employee during the five-day work-week ended at 5 p.m. on Thursday. Pickins faxed the time sheets to the home office, which prepared the payroll checks on Friday mornings. Pickins drove to the home office after lunch every Friday, picked up the payroll checks, and returned to the construction site. At 5 p.m. on Fridays, Pickins distributed the paychecks to the workers.

- a. Describe in detail the internal control weakness in this situation. Specify what negative result could occur because of the internal control weakness.
- b. Describe what you would do to correct the internal control weakness.

L0 3

Case 3. (*Learning Objective 3: Determining allowance for doubtful receivables using the aging method, with and without additional information about loss events*) Two accounting interns, Serene and Joel, were tasked by you, their supervisor, to propose the required amount of

allowance at December 31, 20X7, for Alyssa Candy Empire (ACE), a distributor of specialty confectionery. Data provided to the two interns include an aging schedule below:

Age of Account Receivables					
Receivables	Not Yet Due	1–30 Days	31–60 Days	Over 60 Days	Total
Customer A	400				400
Customer B	100	100			200
Customer C	300	200	600	100	1,200
...	...	...	...	...	...
Totals	11,060	1,363	370	1,093	13,886
Percentage uncollectible	1.0%	5.0%	12.5%	20.0%	
Required allowance.....	111	68	46	219	444

Serene evaluated ACE's historical records of customer defaults and concluded that the likelihood of a receivable becoming bad is correlated to the age of the receivable. She assigned a 1%, 5%, 10%, and 20% likelihood for each age group of receivables.

Joel took another approach and evaluated the likelihood of receivable impairment customer by customer. His research shows that Customer A is a new customer and since it is not yet overdue, there is only a 1% chance that it will not be collected. Customer B and Customer D are long-time customers, and whilst they may pay a little later than the usual credit term of 30 days, the likelihood of not being able to collect their receivables is only 10%. Joel has read that Customer C was not able to make its loan repayments last month. Newspaper articles also point to some worry about Customer C's ability to continue as a growing concern. Joel estimated that it is almost certain that the amount owing would be uncollectible. Customer E, located in another country, has also experienced significant decline in business due to a severe recession in the country. Joel believes that there is a 20% chance that the receivables may be impaired.

Joel and Serene performed their analysis and reported back to you with their recommendations. Whose recommendation will you accept? Why?

Ethical Issue

For each of the following situations, answer the following questions:

1. What is the ethical issue in this situation?
2. What are the alternatives?
3. Who are the stakeholders? What are the possible consequences to each? Analyze from the following standpoints: (a) economic, (b) legal, and (c) ethical.
4. Place yourself in the role of the decision maker. What would you do? How would you justify your decision?

Issue 1. Sunrise Bank recently appointed the accounting firm of Smith, Godfroy, and Hannaford as the bank's auditor. Sunrise quickly became one of Smith, Godfroy, and Hannaford's largest clients. Subject to banking regulations, Sunrise must provide for any expected losses on notes receivable that Sunrise may not collect in full.

During the course of the audit, Smith, Godfroy, and Hannaford determined that three large notes receivable of Sunrise seem questionable. Smith, Godfroy, and Hannaford discussed these loans with Susan Carter, controller of Sunrise. Carter assured the auditors that these notes were good and that the makers of the notes will be able to pay their notes after the economy improves.



Smith, Godfroy, and Hannaford stated that Sunrise must record a loss for a portion of these notes receivable to account for the likelihood that Sunrise may never collect their full amount. Carter objected and threatened to dismiss Smith, Godfroy, and Hannaford if the auditor demands that the bank record the loss. Smith, Godfroy, and Hannaford want to keep Sunrise as a client. In fact, Smith, Godfroy, and Hannaford were counting on the revenue from the Sunrise audit to finance an expansion of the firm.

Issue 2. Barry Galvin is executive vice president of Community Bank. Active in community affairs, Galvin serves on the board of directors of The Salvation Army. The Salvation Army is expanding rapidly and is considering relocating. At a recent meeting, The Salvation Army decided to buy 250 acres of land on the edge of town. The owner of the property is Olga Nadar, a major depositor in Community Bank. Nadar is completing a bitter divorce, and Galvin knows that Nadar is eager to sell her property. In view of Nadar's difficult situation, Galvin believes Nadar would accept a low offer for the land. Realtors have appraised the property at \$3.6 million.

Issue 3. Community Bank has a loan receivable from IMS Chocolates. IMS is six months late in making payments to the bank, and Jan French, a Community Bank vice president, is assisting IMS to restructure its debt.

French learns that IMS is depending on landing a contract with Snicker Foods, another Community Bank client. French also serves as Snicker Foods' loan officer at the bank. In this capacity, French is aware that Snicker is considering bankruptcy. No one else outside Snicker Foods knows this. French has been a great help to IMS, and IMS's owner is counting on French's expertise in loan workouts to advise the company through this difficult process. To help the bank collect on this large loan, French has a strong motivation to alert IMS of Snicker's financial difficulties.

Issue 4. Sunnyvale Loan Company is in the consumer loan business. Sunnyvale borrows from banks and loans out the money at higher interest rates. Sunnyvale's bank requires Sunnyvale to submit quarterly financial statements to keep its line of credit. Sunnyvale's main asset is Accounts Receivable. Therefore, Uncollectible-Account Expense and Allowance for Uncollectible Accounts are important accounts for the company.

	20X7	20X6	20X5
	(In thousands)		
Sales	\$1,475	\$1,001	\$902
Cost of goods sold.....	876	647	605
Gross profit	599	354	297
Other expenses.....	518	287	253
Net profit or (loss) before taxes.....	<u>\$ 81</u>	<u>\$ 67</u>	<u>\$ 44</u>
Accounts receivable.....	\$ 128	\$ 107	\$ 94
Allowance for doubtful accounts.....	13	11	9

Kimberly Burnham, the company's owner, prefers that net income reflect a steady increase in a smooth pattern, rather than increase in some periods and decrease in other periods. To report smoothly increasing net income, Burnham underestimates Uncollectible-Account Expense in some periods. In other periods, Burnham overestimates the expense. She reasons that the income overstatements roughly offset the income understatements over time.

Focus on Financials: | Nestlé



This case spans all 12 chapters and is based on the consolidated financial statements of Nestlé. As you work with Nestlé throughout this course, you will develop the confidence and ability to use the financial statements of other companies as well. Refer to Nestlé's financial statements in Appendix A. If you wish, you can obtain the full annual report from www.nestle.com/investors. You may find the information overwhelming for now, but try to spot the key principles that we have discussed in this chapter. It will get progressively easier as you gain familiarity with the elements of the financial statements.

Requirements

- The bank and cash section of the Balance Sheet shows a balance of CHF 7,990 million as of December 31, 2016, and is made up of many different bank accounts, as well as time deposits, certificates of deposit, and perhaps government securities that are equivalent to cash. Suppose one of Nestlé's year-end bank statements, dated December 31, 2016, has just arrived at company headquarters. Further assume the bank statement shows Nestlé's cash balance at CHF 7,990 million and that Nestlé's record for this bank account has a balance of CHF 8,070 million on the books. You must determine the correct balance for cash in this bank account on December 31, 2016. Suppose you uncover the following reconciling items (all amounts are assumed and are stated in millions):
 - Interest earned on bank balance, CHF 10 million
 - Outstanding checks, CHF 80 million
 - Bank collections of various items, CHF 20 million
 - Deposits in transit, CHF 30 million
 Prepare a bank reconciliation to show how Nestlé arrived at the correct amount of cash in this bank account at December 31, 2016. Journal entries are not required.
- Refer to the account in Nestlé's Balance Sheet entitled "trade and other receivables." This amount is typically shown net of allowances for doubtful accounts. What does "net" mean? How does Nestlé determine its allowance for doubtful accounts? You may refer to Note 7.
- How much is the allowance for bad and doubtful accounts in 2016 and 2015? Assume that the allowance utilized in 2016 was CHF 100 million? Do the journal entries for the writing off of uncollectible accounts and the allowance for bad debts in 2016.

Group Project

You are promoting a rock concert in your area. Assume you organize as a corporation, with each member of your group purchasing \$10,000 of the corporation's shares. Therefore, each of you is risking some hard-earned money on this venture. Assume it is April 1 and that the concert will be performed on June 30. Your promotional activities begin immediately, and ticket sales start on May 1. You expect to sell all of the firm's assets, pay all the liabilities, and distribute all remaining cash to the group members by July 31.

Requirements

Write an internal control manual that will help to safeguard the assets of the business. The manual should address the following aspects of internal control:

- Assign responsibilities among the group members.
- Authorize individuals, including group members and any outsiders that you need to hire, to perform specific jobs.
- Separate duties among the group and any employees.
- Describe all documents needed to account for and safeguard the business's assets.

Quick Check Answers

- | | |
|-------------|--|
| 1. <i>a</i> | 7. <i>b</i> |
| 2. <i>b</i> | 8. <i>a</i> |
| 3. <i>c</i> | 9. <i>b</i> $(\$2,400 - \$90) - (\$400 - \$90)$ |
| 4. <i>a</i> | 10. <i>c</i> $(\$2,800 \times 0.06 \times 4/12 \times 1/4)$ |
| 5. <i>b</i> | 11. <i>b</i> |
| 6. <i>d</i> | 12. <i>b</i> $(365 \times [(\$80,000 + \$74,000)/2] \div \$803,000)$ |

MyLab Accounting

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